

The Link Between Diverse Words and Original Ideas Is Weakening in the AI-Era College Admissions

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Concerns that AI homogenizes human thinking appear at odds with findings that LLM writing is perceived as more creative than human writing. We propose that LLMs enhance surface-level diversity while simultaneously homogenizing underlying ideas. Four large-scale natural experiments tested this preregistered “semantic disjunction” hypothesis in the high-stakes context of college admissions. Comparing personal statements written by 372,793 applicants after-versus-before ChatGPT’s 2022 release, multi-level semantic analyses revealed increasing surface diversity of words simultaneous with decreasing conceptual diversity of sentences (within essays) and whole documents (between essays). Two controlled experiments provided causal evidence that AI use produced semantic disjunction, including in applicants from the natural experiment, and indicated an apparent root mechanism: a positive association between surface and conceptual diversity in humans was weakened or reversed in LLM writing. Despite conceptual homogenization, raters perceived post-ChatGPT essays as more creative. Semantic disjunction explained this apparent paradox: increasing word-level diversity explained increasing creativity judgments, overriding effects of decreased conceptual diversity, even when raters (human and AI) considered many essays together. Avoiding over-reliance on word-level surface diversity by considering multiple semantic levels is likely to support more complete understanding of AI’s effects on human thinking. Disjunctive trends were widespread, but were strongest in minoritized applicants.

As the role of generative AI in human creative processes rapidly expands¹, evidence of LLM-driven homogenization, broadly defined as reduced diversity in LLM-generated²⁻⁷ or LLM-influenced outputs⁸⁻¹¹, has raised concerns that LLMs may diminish human idea diversity¹²⁻¹⁴. This homogenization largely reflects the probabilistic architecture of LLMs: by predicting the most probable next words (tokens), LLMs overrepresent normative linguistic and conceptual patterns embedded in the texts on which they are trained^{12,15-19}. Reinforcement Learning from Human Feedback (RLHF) further intensifies this tendency by rewarding safe, conventional responses²⁰⁻²³. Because originality is a defining feature of creative outputs²⁴⁻²⁷, limited diversity in LLM outputs might be expected to reduce their perceived creativity. Paradoxically, however, AI-generated and assisted content is frequently rated as equally or even more creative than content produced by humans despite greater similarities across outputs^{2-11,28,29}. How can perceived creativity rise as originality falls?

One plausible explanation for this paradox is an AI-driven disjunction between surface-level lexical features and underlying conceptual content. When people judge the creativity of a text, perceived creativity is positively associated with surface variations (e.g., unique word count and lexical/semantic diversity)³⁰⁻³³, even when conceptual novelty does not differ^{6,34}. For readers of human writing, this reliance on surface linguistic cues has generally not been problematic because greater lexical diversity is a reasonable proxy for deeper conceptual diversity. Vocabulary size is linked to creativity, and this relationship appears to stem from the largely shared neurocognitive bases of verbal ability and creativity^{35,36}. The flexible use of language is closely related to creative thinking, such that greater lexical richness, a more flexible semantic network structure, and higher retrieval flexibility all support the generation of diverse and original ideas by facilitating more remote conceptual associations³⁶⁻⁴¹. In the architecture of LLMs, however, these mechanisms of human cognition that link surface variability of words

with conceptual originality do not appear relevant. Because LLM generation relies on stochastic sampling from learned distributions rather than a search through an internally structured semantic network, surface features can be optimized independently of the underlying conceptual structure in AI text^{16,18,40}.

5 Thus, a fundamental question is whether lexical and conceptual semantic levels are less related in LLMs than in humans. If LLMs break this link—more readily generating lexically rich language that nonetheless conveys probabilistically homogenized ideas^{14,15,42}—surface-level semantic diversity would be an unreliable indicator of substantive originality. LLM-influenced content, therefore, might exhibit a “semantic disjunction,” appearing more creative on the
10 surface while actually reducing conceptual originality.

 Testing semantic disjunction requires assessing semantic diversity at multiple levels simultaneously: word, sentence, and whole-document level. At the word level, semantic diversity captures variation of vocabulary use within each text. At the sentence level, semantic diversity captures the local variation of ideas within each text. At the document level, semantic diversity
15 captures the global distinctness of ideas in an entire text from other texts in a corpus. Thus, word-level diversity is an indicator of surface-level semantic diversity, whereas sentence-level diversity and especially document-level distinctness are indicators of deeper conceptual diversity (see Methods for operational definitions, methodological details, and supporting literature for each metric).

20 Comparing the effects of LLMs on these three levels of semantic diversity makes it possible to 1) observe whether surface and conceptual semantic features diverge when LLMs are introduced, 2) investigate whether LLMs break the link between word diversity and conceptual originality that is characteristic of human writing, and 3) compare the influences of surface vs. conceptual semantic features on judgments of creativity in the context of AI-driven changes. If

reliance on surface-level diversity of words rather than underlying diversity of concepts contributes to high creativity ratings of LLM-generated and LLM-influenced text, this would implicate semantic disjunction as an explanation for the apparent paradox of increasing creativity ratings in the AI era. In particular, if increased word-level diversity overrides decreased document-level distinctness even when raters evaluate many essays together, this would challenge accounts that attribute this paradox to individual evaluation of texts without exposure to group-level similarities^{2,7,8,43}.

In the present work, across four large-scale natural experiments and two controlled experiments, we examined the influences of LLM availability and use on the semantic properties of writing and creativity assessment at both surface and conceptual levels in the context of personal statement college admissions essays. This context was of particular interest because a primary objective of this research was to test the extent of the hypothesized LLM-era changes in real-world human writing. Personal statement admissions essays have several characteristics that make for a stringent and instructive test. First, the writing, including any interactions with AI, occurs in fully naturalistic contexts, unlike laboratory-based or remote experiments in which AI use is instructed and often constrained. Second, the writing is high-stakes because it can influence admissions outcomes that are frequently of high value to applicants. High stakes are likely to motivate applicants to strategically leverage LLMs to optimize their writing (e.g., investing effort in extensive revision), which can be difficult to capture in lower-stakes contexts⁴⁴. Third, the use of AI in these essays is widely prohibited by application platforms such as the Common Application⁴⁵ and many colleges⁴⁶. This, in combination with the high stakes, is an ostensible barrier against AI-era changes that appear in non-transgressive and/or lower-stakes writing. Fourth, personal statement essays are a useful source of data for assessing creativity. Creativity and individual expression are frequently encouraged in personal statement essay

prompts, including in this study's data set, and creativity is consistently valued by admissions offices in evaluating applicants⁴⁷⁻⁴⁹. Admissions essays have previously provided a useful basis for assessing creativity in both laboratory and applied contexts^{47,50-54}.

Finally, prompts for these essays ask applicants to communicate their personal experiences, perspectives, and qualities that make them uniquely themselves. Responses are often highly personal in nature. These "life story" narratives are valuable for assessing putative disparities in the impacts of LLMs across racial and linguistic demographics^{15-17,44}. Previous scholarship indicates that the experiences and lexical norms of minoritized demographic groups are likely to be less represented in LLM training data, subsequent RLHF processes, and ultimately the probabilistic predictions these models make^{15-18,23,42,55}. This suggests the hypothesis that the personal narratives of minoritized applicants may be reshaped more by the advent of LLMs (on surface and conceptual levels) than the narratives of their White-identifying counterparts.

We conducted four natural experiments contrasting personal statement admissions essays submitted to four U.S. universities before and after the 2022 release of ChatGPT. These four universities included a flagship state/public university, an elite private university, a regional private university, and a selective liberal-arts college, reflecting variations in selectivity, geography, size, and applicant pool composition. The natural experiments used two related samples. The primary balanced analytic sample included 372,793 essays with equalized cycle-level sample sizes within each institution to support document-level distinctness comparisons (see Methods and Supplementary Table 1a for details). The full unbalanced dataset included 394,199 essays across the same institutions and was used for robustness checks. Institution-specific coverage, exclusions, demographic composition, essay-length descriptives, and prompt distributions are reported in Supplementary Tables 1b-5, and Extended Data Fig. 1. The

application window of the 2023 admissions/matriculation cycle overlapped ChatGPT's November 30, 2022, release and was therefore excluded from the primary pre/post analyses as a transition cycle (see Methods for details).

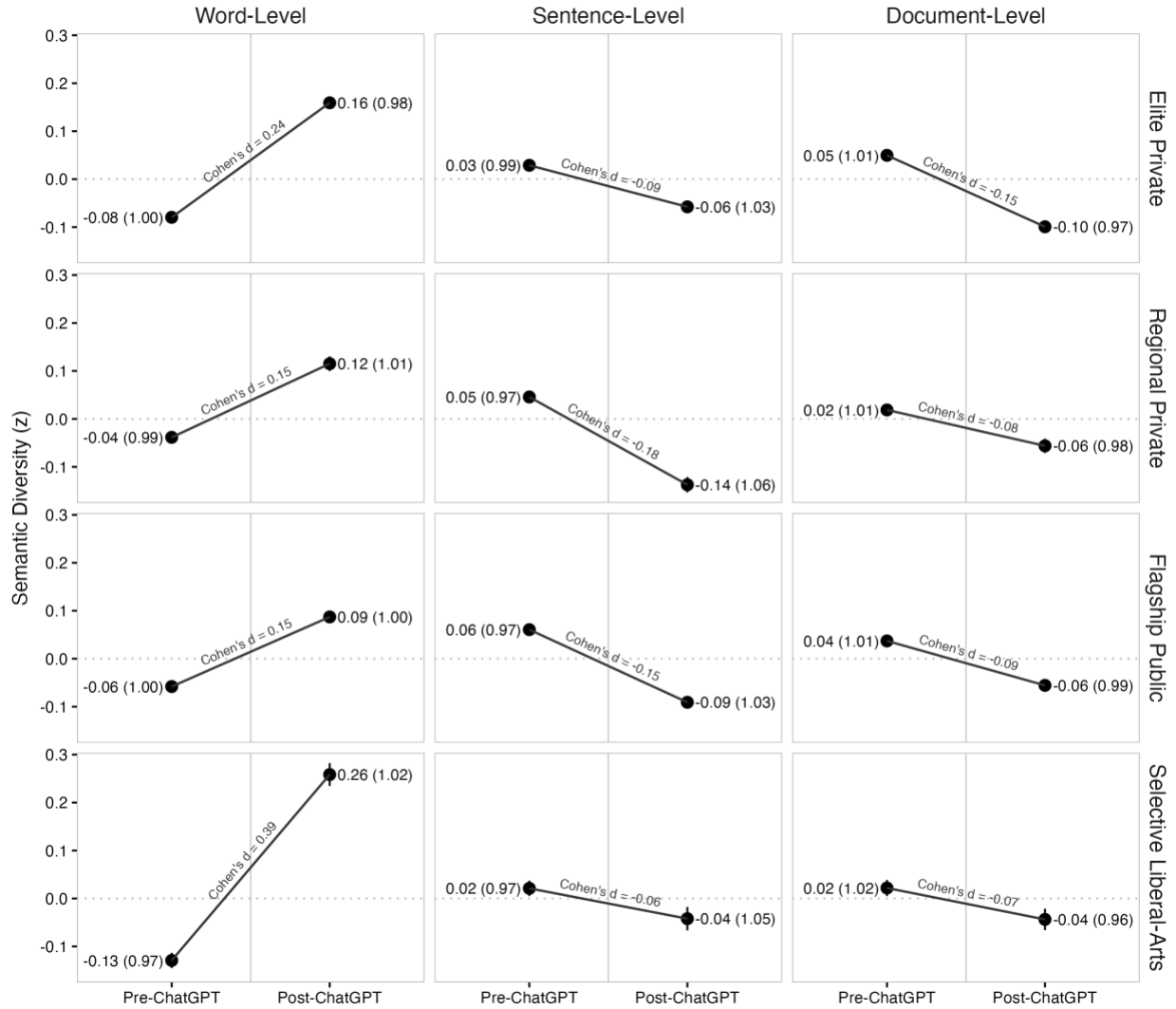
To distinguish surface-level semantic diversity from the conceptual diversity of underlying ideas, we applied a multi-level semantic analysis at the word, sentence, and document levels (see Methods and Table 3). Building on an initial pilot study (Supplementary Table 6), we predicted that post-ChatGPT personal statements would exhibit a *semantic disjunction* pattern, characterized by enhanced surface-level semantic diversity but a countervailing conceptual convergence both within essays (sentence-level) and between essays (document-level). We also explored how enhanced surface-level semantic diversity was related to the creativity of LLM-influenced text as perceived by raters, even when raters considered multiple essays. Finally, we examined the semantic disjunction effect in a controlled within-subjects experiment to test the causal role of AI in the effects observed in our natural experiments. Summaries of all hypotheses, preregistrations, and outcome details are reported in Extended Data Table 1. Full preregistration documents for all four institutions are available on the Open Science Framework at <https://osf.io/yd94z>.

Results

Natural Experiments

At each of the four universities, the multi-level semantic analysis showed a pattern consistent with our preregistered hypothesis of semantic disjunction (Fig. 1 and Supplementary Table 7; see also *Natural Experiments* in Methods for a full description of this analysis). We found that post-ChatGPT essays showed greater word-level diversity than pre-ChatGPT essays, indicating enhanced lexical variation (Cohen's d : 0.15 to 0.39, $ps < .001$). Simultaneously, post-ChatGPT essays showed lower sentence-level diversity (Cohen's d : -0.18 to -0.06, $ps < .001$) and

document-level distinctness (Cohen's d : -0.15 to -0.07, $ps < .001$), indicating more homogeneous ideas within and across essays. In short, we found that the more semantically diverse words in post-ChatGPT essays were conveying more homogeneous ideas.



5 **Fig. 1.** Natural Experiments: Semantic Diversity Changes in College Admissions Essays (Pre- and Post-ChatGPT Differences). Semantic diversity scores were standardized within each semantic level using pooled data across all cycles separately for each institution, excluding the 2023 transition cycle. Data points represent the mean of standardized scores (z), with standard deviations provided in parentheses. Error bars represent 95% confidence intervals around the

means. All Welch's t -tests comparing standardized semantic diversity scores between pre- and post-ChatGPT essays were significant ($p < .001$). Sample periods and sizes varied across institutions: elite private – pre: 2019–2022 ($n = 47,520$), post: 2024–2025 ($n = 23,760$); regional private – pre: 2020–2022 ($n = 49,785$), post: 2024 ($n = 16,595$); flagship public – pre: 2020–
5 2022 ($n = 92,790$), post: 2024–2025 ($n = 61,860$); selective liberal-arts – pre: 2018–2021 ($n = 14,052$), post: 2024–2025 ($n = 7,026$). See Supplementary Table 7 for full results and Supplementary Table 5 for results based on the full, unbalanced sample ($N = 394,199$ across four universities).

To assess whether the post-ChatGPT semantic shifts reflected AI-related changes, we conducted a diagnostic analysis using GPTZero, an independent AI detector⁵⁶. In a balanced sample of 2,000 anonymized essays from four universities, GPTZero classified 1 of 1,000 pre-ChatGPT essays and 125 of 1,000 post-ChatGPT essays as AI or mixed. These rates likely underestimate actual AI use because detector false negatives are expected in admissions contexts. We therefore treated the detector classifications as a conservative diagnostic signal. We found that post-ChatGPT essays classified as AI or mixed reproduced the pattern of Semantic Disjunction observed in the main natural experiment data sets with substantially larger absolute effect sizes across institutions: on average, the word-level increase was 1.9-fold larger than in the main natural experiment, and the sentence-level and document-level decreases were 5.3-fold and 5.8-fold larger, respectively (see *Diagnostic Analysis* in Supplementary Information; Extended Data Fig. 2 and Supplementary Table 8).

We conducted additional robustness checks to rule out alternative explanations. First, event-time analyses confirmed that the conceptual homogenization did not simply reflect a pre-existing trend (see *Event-Time Analysis* in Supplementary Information; Extended Data Fig. 3 and Supplementary Tables 9a–c). Second, this pattern remained after accounting for potential confounding effects of the COVID-19 pandemic (see *COVID-19 Pandemic Confound Analysis* in Supplementary Information; Supplementary Tables 10a–d). Third, the observed patterns of semantic disjunction remained significant after controlling for variables such as essay length, sex, race and ethnicity, international status, and home language, where available (see *Models with Control Variables* in Supplementary Information; Supplementary Tables 11a–d). Finally, the word-level increase was also observed in a between-essay analysis, indicating that the disjunction between word-level diversity and document-level distinctness was not an artifact of

comparing within-essay and between-essay measures (see *Collective Word-Level Diversity Across Essays* in Supplementary Information; Extended Data Fig. 4).

Racial and Linguistic Subgroup Analyses

We found a consistent pattern across universities for most demographic groups, supporting the generalizability of semantic disjunction (see *Racial and Linguistic Subgroup Analyses* in Supplementary Information for a full description of this analysis). As shown in Fig. 2, the differences in the estimated marginal means between pre- and post-ChatGPT were directionally consistent across subgroups, with nearly all subgroups exhibiting the pattern of opposing trends across semantic levels: positive shifts at the word level and negative shifts at the sentence and document levels.

The magnitude of the effect was more pronounced among applicants from marginalized and non-mainstream backgrounds. Compared with the White American reference group, Black/African American applicants, international applicants, and applicants from non-English-speaking households frequently exhibited larger increases in word-level diversity alongside sharper declines in conceptual diversity at the sentence and document levels (Supplementary Tables 12a–e and Extended Data Fig. 5). The selective liberal-arts college had the smallest analytic sample, making subgroup-specific document-level estimates less stable. However, when subgroups were pooled, the effects were highly comparable to those observed at the other universities. In the corresponding full-sample analysis, all document-level estimates across subgroups were negative, consistent with the broader post-ChatGPT decline in document-level distinctness (Extended Data Fig. 1).

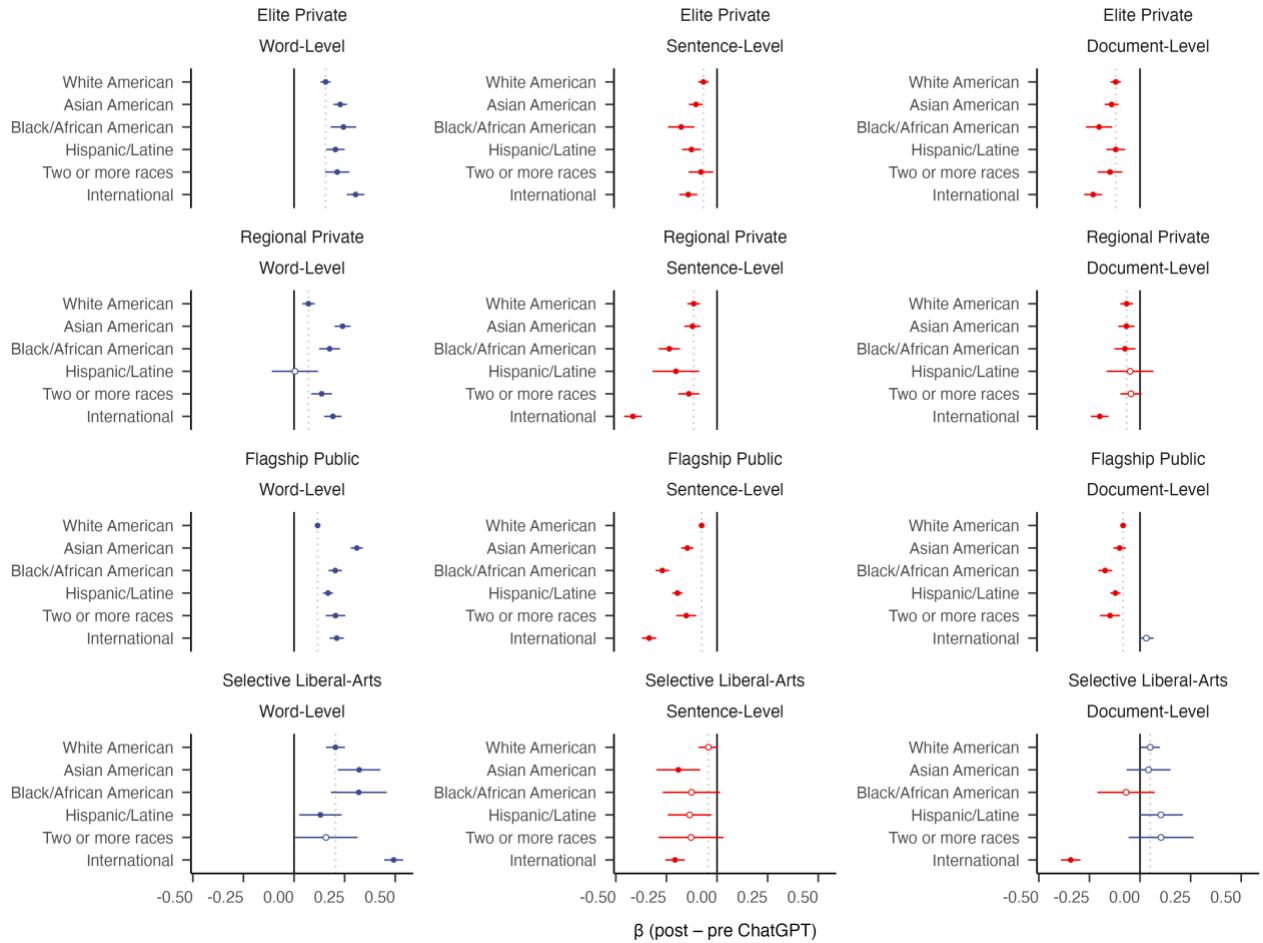


Fig. 2. Natural Experiments: Generalizability of Semantic Disjunction Across Racial and Ethnic Subgroups. Forest plots display the estimated change in semantic diversity (β) following the release of ChatGPT (post-ChatGPT minus pre-ChatGPT) across four U.S. universities. Dots represent point estimates and error bars indicate 95% confidence intervals. Solid dots indicate statistically significant effects, after Holm correction for multiple comparisons within each semantic level. Hollow dots indicate non-significant effects. Grey dotted lines indicate the effect of the White American applicants within each university and semantic level. Applicants with Indigenous backgrounds or with missing/undisclosed demographic information were included in the models but were omitted from the plots due to small subsample sizes. See Extended Data Fig. 1 for results based on the full, unbalanced sample ($N = 394,199$ across four universities).

Investigating Mechanisms

To investigate a potential root cause of AI's apparent semantic disjunction in essay writing, we explored whether LLM writing decouples surface semantic diversification from underlying idea diversity and conceptual novelty. We conducted an experiment comparing human-written ($n = 1,500$) and GPT-generated essays ($n = 1,500$) matched on essay prompt choice (see *Controlled Experiment 1* in Supplementary Information for details). We found that human-written essays exhibited positive associations across all three semantic levels, whereas LLM-generated essays exhibited a negative association between the word- and document-level measures. Specifically, word-level diversity positively correlated with document-level distinctness in human writing ($r = .117, p < .001$) but showed the opposite pattern in AI-generated text ($r = -.283, p < .001$). As shown in Supplementary Table 13a, human essays also exhibited a stronger positive association between word- and sentence-level diversity ($r = .430, p < .001$) than AI-generated text ($r = .051, p < .05$). We replicated this analysis in the natural experiments comparing pre- and post-ChatGPT essays and found results that were consistent with this pattern for the decoupling of word and document levels across each of the four universities (Supplementary Tables 13b–c). Decoupling of word and sentence levels was less consistent, emerging for two of the four universities.

Resolving Paradoxical Effects

How does semantic disjunction impact perceptions of the creativity of essays? Do raters rely on surface-level word diversity when judging essay creativity, or do they perceive the underlying conceptual diversity? To examine these questions, it was necessary to obtain ratings for creativity, as distinct from the multi-level quantitative measurements of semantic diversity described for the other experiments, so that creativity ratings could be related to those levels of semantic diversity. We thus started by using a separate dataset of $N = 370$ human-written essays

rated by 22 experts specializing in creativity research (see *Resolving Paradoxical Effects* in Supplementary Information for details). In a multiple regression predicting ratings from word-, sentence-, and document-level metrics, we found that word-level diversity emerged as the strongest predictor of expert creativity ratings ($\beta = 0.42, p < .001$), significantly exceeding the influence of both sentence-level diversity ($\beta = 0.20, p < .001$) and document-level distinctness ($\beta = 0.22, p < .001$). This surface-level lexical bias was confirmed by tests comparing the effect sizes of word- with sentence- and document-level diversity [$F_s(1, 366) \geq 9.02, p_s \leq .003$].

Notably, this lexical bias existed even though the human experts were explicitly asked to read approximately 50 essays before providing creativity ratings and to rate each essay in the context of all the essays they read. In short, human experts relied more on surface-level semantic diversity than on underlying idea diversity and thematic distinctness, suggesting that semantic disjunction may be associated with higher creativity ratings despite lower conceptual diversity.

To directly evaluate how semantic disjunction impacts perceptions of creativity across our large natural experiments, we developed an automated evaluator by fine-tuning GPT-4.1-mini on the same expert-rated human essays ($N = 370$). As shown in Table 1, this expert-tuned model demonstrated high validity, strongly correlating with human expert ratings on the held-out test set ($r = .82, p < .001$) and outperforming the base GPT-5-mini model ($r = .67, p < .001$; see *Resolving Paradoxical Effects* in Supplementary Information for full fine-tuning procedures). As shown in Table 2, the model mirrored the surface-level lexical bias inherent in the expert ratings, and tests comparing the effect sizes confirmed this bias in model ratings [$F_s(1, 107) \geq 5.54, p_s \leq .020$]. This lexical bias persisted even after asking the model to rate the creativity of each essay in the context of all other essays. Specifically, we found no systematic effects on the relative role of word-, sentence-, and document-level metrics when experimentally increasing the number of

example essays (from 0 to 128) in the evaluation prompt (see *Controlled Experiment 2* in Supplementary Information and Extended Data Fig. 6).

	(1)	(2)	(3)	(4)	(5)	(6)
1. Word-Level	–					
2. Sentence-Level	.43***	–				
3. Document-Level	.24***	.31***	–			
4. Human Expert Ratings	.56***	.45***	.39***	–		
5. Expert-Tuned GPT-4.1-mini	.63***	.48***	.44***	.82***	–	
6. Base GPT-5-mini	.58***	.39***	.45***	.67***	.67***	–
Mean (SD)	0.84 (0.01)	0.76 (0.05)	0.28 (0.07)	2.96 (1.02)	0.44 (0.26)	0.34 (0.16)

Table 1. Assessment Analysis: Correlations between Semantic Metrics and Creativity Ratings. $N = 370$ ($n = 111$; only the held-out test dataset was used for GPT models). *** $p < .001$.

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	Human Expert Ratings	Expert-Tuned GPT-4.1-mini
	β	β
Word-Level	0.42***	0.49***
Sentence-Level	0.20***	0.22**
Document-Level	0.22***	0.22**
N	370	111
R^2	.415	.517

Table 2. Assessment Analysis: Multiple Regression Models Predicting Creativity Ratings from Semantic Metrics. ** $p < .01$. *** $p < .001$.

Next, we used the expert-tuned GPT-4.1-mini model to examine how semantic disjunction in post-ChatGPT essays impacted creativity ratings. Consistent with previous findings that both AI-generated and AI-assisted human outputs can be evaluated as more creative or original^{2,3,6,9,10,28}, post-ChatGPT essays received higher creativity ratings than pre-ChatGPT essays across all four universities (total effect = 0.19 to 0.30, $ps < .001$; see Supplementary Table 14). To explore whether these perceptions of higher creativity in post-ChatGPT essays were due to surface-level lexical bias, we tested a parallel mediation model with word-, sentence-, and document-level diversity as mediators (Fig. 3). Each model was just-identified ($\chi^2(0) = 0$, $p = 1.00$) so no fit indices were interpreted. We estimated all direct, indirect, and total effects using bootstrapping with 5,000 resamples to generate confidence intervals.

We found a positive indirect effect of post-ChatGPT essays on creativity ratings via word-level diversity ($a_{\text{word}} \cdot b_{\text{word}} = 0.08$ to 0.21 , $ps < .001$) and simultaneous negative indirect effects via sentence and document levels (sentence: $a_{\text{sent}} \cdot b_{\text{sent}} = -0.02$ to -0.01 , $ps < .001$; document: $a_{\text{doc}} \cdot b_{\text{doc}} = -0.01$ to -0.00 , $ps < .001$). Bootstrapped contrasts showed that the word-level indirect effect was larger than the combined sentence- and document-level effects ($\text{diff} = 0.06$ to 0.20 , $ps < .001$). Accounting for these three indirect paths results in the attenuated direct effect ($c' = 0.08$ to 0.18 , $ps < .001$), indicating partial mediation. These findings indicate that increased surface diversity of words in post-ChatGPT writing accounted for increased creativity ratings, even as deeper conceptual levels became more homogeneous.

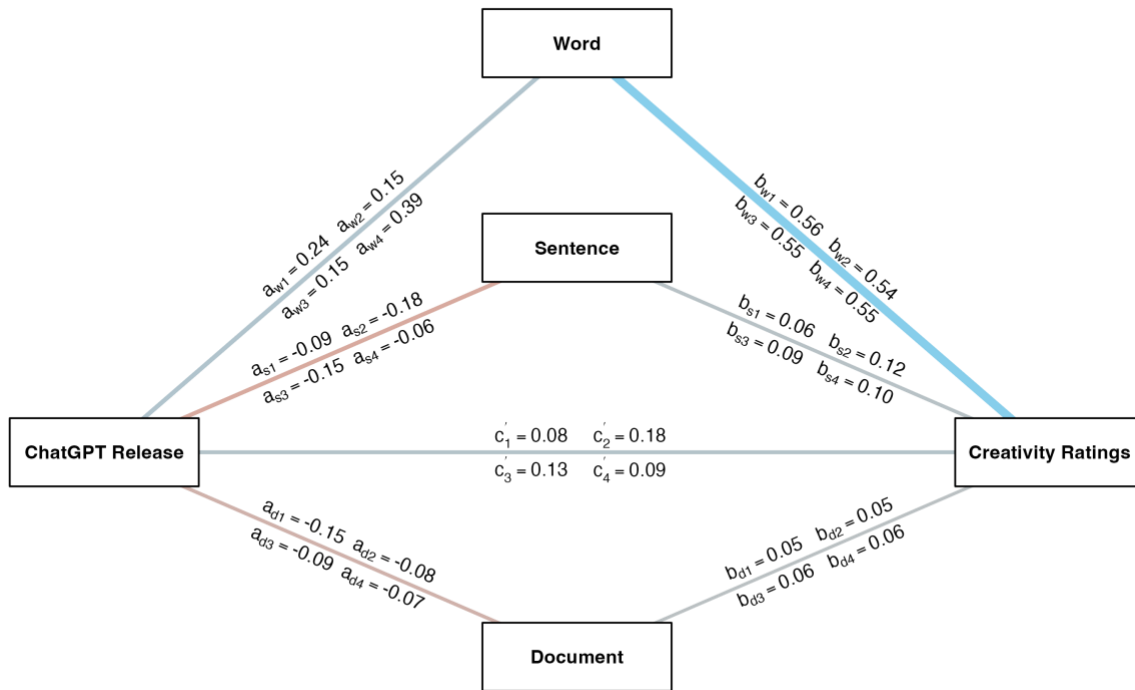


Fig. 3. Natural Experiments: Parallel Mediation of ChatGPT Release Effects on Creativity

Ratings via Three Semantic Layers. “ChatGPT Release” denotes a binary post- vs. pre-release indicator. “Word,” “Sentence,” and “Document” refer to semantic metrics at each level.

Creativity ratings were produced by the GPT-4.1-mini model fine-tuned on human expert ratings. We separately estimated a parallel mediation model for each institution and presented the institution-specific standardized path coefficients around each path (elite private1 and regional private2 shown above the path; flagship public3 and selective liberal-arts college4). All paths were significant at $p < .001$. For descriptive visualization, we fit the same model to the pooled sample across all four universities and used the standardized coefficients to set line color (blue = positive, red = negative) and line width (proportional to magnitude). See Supplementary Table 14 for full results.

Testing Semantic Disjunction Within-Subjects

To test the causal influence of AI use on semantic diversity at the surface (word) and conceptual (sentence and document) levels, we conducted a within-subjects experiment. College students ($N = 203$) were recruited from the same elite private university as in the natural experiments and asked to upload their college admissions essays, which they had originally submitted when applying to the college they attended. Each participant then created two ChatGPT prompts: one for revising their original essays with AI and another for generating a completely new essay using AI. Importantly, we encouraged participants to design prompts that would help preserve their individual voice and originality. We then used these prompts to produce AI-revised and AI-generated essays using GPT-4.1 to compare with the original essays (see *Controlled Experiment 3* in Supplementary Information for details).

We compared each participant's semantic diversity scores across (1) original essay, (2) AI-revised essay, and (3) AI-generated essay using linear mixed-effects (multilevel) models with random intercepts for participants to account for repeated measures. Consistent with the patterns of the natural experiments, we found evidence for semantic disjunction as a result of AI use (Fig. 4 and Supplementary Table 15a). At the word level, both AI-revised essays ($diff = 0.79$, $p_{Holm} < .001$, 95% CI [0.64, 0.94]) and AI-generated essays ($diff = 0.65$, $p_{Holm} < .001$, 95% CI [0.50, 0.79]) showed a significant increase in word-level diversity compared to original essays. In contrast, AI use significantly decreased sentence-level diversity for both AI-revised essays ($diff = -0.47$, $p_{Holm} < .001$, 95% CI [-0.62, -0.31]) and AI-generated essays ($diff = -0.50$, $p_{Holm} < .001$, 95% CI [-0.66, -0.35]), indicating that ideas *within* each essay became more homogeneous. Likewise, AI use significantly decreased document-level distinctness for both AI-revised essays ($diff = -0.98$, $p_{Holm} < .001$, 95% CI [-1.10, -0.85]) and AI-generated essays ($diff = -1.52$, $p_{Holm} < .001$, 95% CI [-1.64, -1.40]), indicating greater homogenization of ideas *across* essays.

Differences between the two AI conditions were not significant at the word or sentence levels, but document-level distinctness was greater in AI-revised essays than in AI-generated essays ($diff = 0.55$, $p_{Holm} < .001$, 95% CI [0.39, 0.70]), suggesting a relative advantage of human–AI collaboration compared to AI-only creation in preserving global conceptual distinctness. These effects were not significantly moderated by participants’ prior experience using LLMs, $\chi^2s(2) \leq 1.61$, $ps \geq .447$ (Supplementary Table 15b).

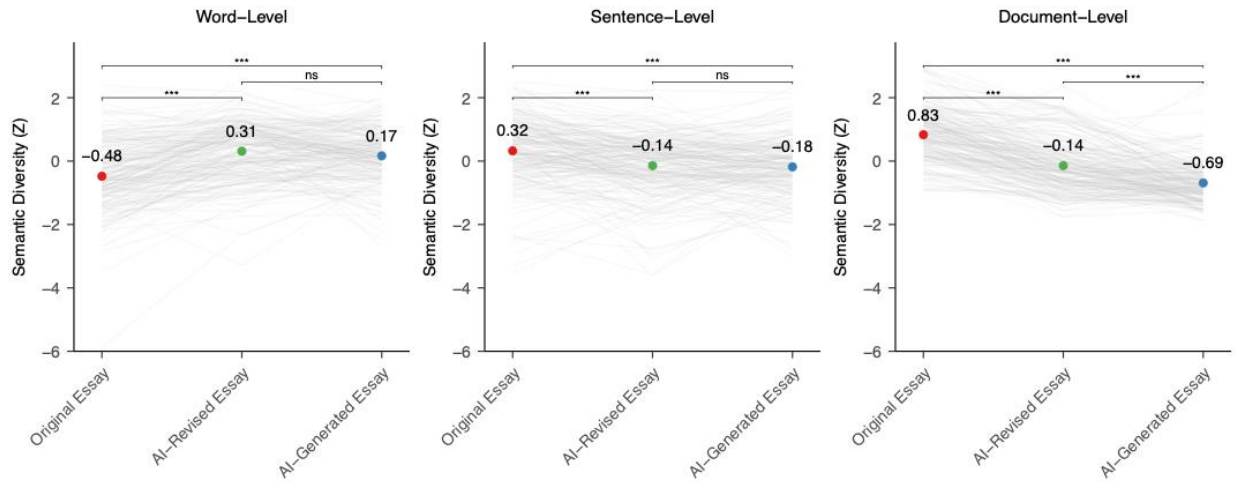


Fig. 4. Controlled Experiment 3: Within-Subject Effects of AI Use on Semantic Diversity. $N = 609$ essays (203 participants $\times$ 3 writing conditions) per semantic level. Spaghetti plots trace within-participant variation in each semantic level across writing conditions. Dots and error bars show estimated marginal means and 95% confidence intervals from mixed-effects models for each authorship condition (Supplementary Table 15a). Post-hoc pairwise comparisons were used to compare estimated marginal means with Holm correction for multiple comparisons, conducted separately for each semantic level. *** $p < .001$. ns = not significant.

Discussion

Four large-scale natural experiments in the high-stakes context of college admissions essay writing found an emerging disjunction between words and ideas in the AI era (increasingly diverse words conveyed decreasingly distinct ideas). This disjunction explained a paradoxical effect that is characteristic of AI-era content in which judgments of creativity increased even as ideas became more homogeneous. Subsequent controlled experiments provided evidence for AI's causal role and identified an LLM reversal of word-idea links that characterize human writing as a plausible root mechanism.

The preregistered natural experiments in four U.S. universities, representing a range of institutional and applicant characteristics, found that personal statement admissions essays written after the release of ChatGPT became significantly more diverse at the word level. However, this surface-level diversification masked a deeper conceptual convergence: the underlying ideas within each essay became more homogeneous, and global/thematic ideas across essays became less distinct. We conceptually replicated this pattern of semantic disjunction in a controlled experiment, providing evidence for the causal role of LLM use: both AI-revised and AI-generated essays showed increased word-level diversity alongside reduced sentence-level diversity and document-level distinctness relative to original essays. Notably, however, AI-revised essays retained greater document-level distinctness than fully AI-generated essays. This suggests that human–AI collaboration may partially preserve thematic uniqueness compared to AI-only generation, even though both forms of AI use contribute to semantic disjunction.

Despite their lower conceptual diversity, post-ChatGPT essays were rated as more creative than pre-ChatGPT essays. This increase in perceived creativity was explained by a strong word-level bias in human and fine-tuned AI raters: mediation analyses indicated that the increase in creativity ratings of post-ChatGPT essays was primarily explained by their increased

surface-level diversity of words, outweighing the negative indirect effects of conceptual homogenization on the judgments of raters. Importantly, this word-level bias was consistent even when humans and a fine-tuned model considered multiple essays jointly. This suggests that mere exposure to multiple texts may be insufficient to reduce emphasis on surface cues in creativity assessment. Semantic disjunction may thus explain the ostensibly paradoxical phenomenon of AI-generated and AI-assisted writing being judged as highly creative despite evidence of homogenization^{2,3,6,9,10,29}. This explanation challenges the completeness of prior accounts of this paradox focused on the limits inherent in single evaluation^{2,7,8,43}; even under joint evaluation, word-level variation continued to dominate creativity judgments. This is consistent with prior (pre-LLM era) evidence that human raters neglect conceptual distinctness even in joint evaluation⁵⁷.

A direct comparison of LLM-written essays to human-written essays indicated a plausible root mechanism for semantic disjunction. In human writing, greater word-level diversity was positively associated with diversity at conceptual levels. This is consistent with previous evidence linking a more varied vocabulary with flexible semantic memory structure and creative idea generation^{35–40}. In contrast, our analyses showed that this relationship was substantially weakened—or even reversed—in LLM-generated text and in post-ChatGPT admissions essays. Essays exhibiting higher word-level diversity were often less thematically distinct from one another. Breaking the human-characteristic link between words and ideas is likely a result of LLM architectures that readily vary lexical choices through stochastic generation over learned language distributions without a corresponding expansion of the conceptual space those choices express, enabling surface linguistic variation to increase even as underlying ideas become more homogenized.

The socio-demographic implications of semantic disjunction in our large-scale data set appear to be both widespread and uneven. Although the pattern was observed across nearly all applicant subgroups, it was more pronounced among applicants from marginalized and non-mainstream backgrounds. International applicants, Black/African American applicants, and students from non-English-speaking households showed larger increases in word-level diversity alongside steeper declines in sentence-level and document-level diversity. One possibility is that these applicants face greater pressure to conform to dominant linguistic and cultural norms embedded in LLMs, which may disproportionately reshape their narratives toward more normative themes. In this way, AI tools may unintentionally dampen the distinctness of the marginalized personal perspectives that institutions often value in efforts to build diverse student communities.

One limitation of the natural experiments is that causal inference depends on the assumption that pre- and post-ChatGPT applicant cohorts are broadly comparable on unobserved characteristics. As discussed above, this concern was mitigated by triangulating evidence across multiple institutions with distinct applicant pools and prompts, AI-detector diagnostics, robustness checks, and complementing the natural experiments with the controlled within-subjects design. The convergence of these approaches strengthens confidence that the observed pattern reflects AI-related change rather than coincidental cohort differences. Nonetheless, some unmeasured contextual changes may remain. Further, while measuring semantic distinctness within a corpus of texts is consistent with long-standing and current methods for assessing originality^{32,58,59}, it should be noted that this approach does not allow inferences about historical newness of ideas because essays were compared within yearly cohorts. It should also be noted that word-level diversity showed a modest upward trend prior to the release of ChatGPT. While results of the controlled experiment indicated that AI use causally increased word-level diversity,

this pre-existing trend suggests that lexical diversification may also be related to other factors. The influence of pre-ChatGPT AI writing tools (e.g., Grammarly) introduced during these years is one possibility.

Although the primary target of the present study was quantitative assessments of semantic diversity at multiple levels, one of the controlled experiments also involved human and AI ratings of creativity. It should thus be noted that personal statement admissions essays are not designed specifically for creativity assessment, e.g., applicants' primary objective is generally persuasion rather than creativity, even if colleges encourage creativity. Nonetheless, because the present study focused on changes over time across the population of essays, and the relationships of different semantic levels of diversity (words vs. ideas) to the assessment of creativity, these essays appear to provide an adequate signal. Creativity experts' ratings indicated that creativity could readily be detected (see *Resolving Paradoxical Effects* in Supplementary Information), consistent with prior assessments of creativity and originality in applicant essays and other student essays with persuasive (rather than primarily creative) objectives^{51–54,60,61}. The presence of non-creativity objectives for many students (which is likely to be a constant factor across time) appears unlikely to systematically influence changes in the creativity that is present in the population of essays over time or the relationships of different semantic levels to perceived creativity in the essays.

The emergence of post-ChatGPT semantic disjunction trends in personal statement admissions essays occurred in the context of ostensible boundaries to AI influence, including AI prohibition, high-stakes evaluation, incentives to demonstrate individuality, and the frequently personal nature of the writing. Our findings thus suggest that the reach of these disjunctive trends should be considered across AI-era writing and communication more broadly, including other consequential real-world contexts^{62–64}. Semantic disjunction appears to represent a new

separation of language elements that have been linked in human writing. Relatedly, semantic disjunction challenges a long-standing evaluative heuristic: in human writing, lexical diversity has served as a useful signal of conceptual originality, but the present findings suggest that this surface signal is becoming disjoined from conceptual originality in LLM-influenced writing.

5 Simply considering many texts together appears to be insufficient to restore the reliability of this heuristic. Assessing multiple semantic levels simultaneously, and especially avoiding over-reliance on word-level diversity, is likely to support a more complete understanding of AI's effects on writing. This may be especially important when comparing human-generated vs. AI-influenced content or when evaluating claims that AI assistance improves human originality and
10 creativity – currently, both human and AI approaches to evaluating such claims appear likely to anchor on word-level cues that may indicate improvement even as the originality of ideas declines.

Methods

Ethical Compliance

15 This research was approved by the institutional review board of University X (natural experiments 1, 2, 3, and 4: *[Redacted]*; controlled experiment: *[Redacted]*) and complied with all relevant ethical regulations.

Semantic Diversity Metrics

To quantify semantic disjunction trends, we compared the semantic diversity of text at three
20 semantic levels, varying embedding granularities at word, sentence, and document levels. At each level, we used a computational metric based on semantic distance, which has been widely used in previous research^{32,65,66}. At its core, this involves transforming words, sentences, or documents into numerical representations, known as text embeddings, which capture their

underlying semantics. Semantic distance then refers to how close or far apart these embeddings are in a high-dimensional space; texts that are closely associated will have a smaller semantic distance, whereas semantically unrelated pairs will have a bigger one. By analyzing these distances, we can assess the overall semantic diversity at each level; a larger overall distance suggests greater diversity in concepts and ideas, whereas a narrower overall distance indicates higher homogeneity^{32,67}. Detailed information for each semantic diversity measure is provided below (see also Table 3).

Word-Level Diversity. For word-level diversity, we utilized the Divergent Semantic Integration (DSI)³². DSI was originally designed to assess how well a text connects divergent ideas. It is calculated by averaging the semantic distances between all word pairs within a text, using word-level BERT embeddings retrieved from middle layers (layers 6 and 7) of the *bert-large-uncased* model⁶⁸. DSI shows strong validity, with correlations to human experts' creativity ratings ($rs = .73-.85$) across various creative story-writing tasks³². In our study, higher DSI scores indicate greater surface-level semantic diversity.

Sentence-Level Diversity. For sentence-level diversity, we computed the mean pairwise distance between sentence embeddings, following Anderson et al.⁸ Specifically, we used sentence embeddings generated by the *all-MiniLM-L6-v2* model, a variant of the Sentence-BERT (SBERT) models⁶⁹. SBERT is a modification of BERT, fine-tuned using Siamese or Triplet network structures to generate semantically meaningful sentence embeddings⁶⁹. As with word-level embeddings, sentence-level embeddings were used to compute the semantic distance between two sentences based on cosine distances. The mean pairwise distance between sentence embeddings has been reported as a reliable measure of semantic homogenization⁸. A study found that the cosine distances from SBERT embeddings aligned more closely with human judgments of semantic similarity than those from GloVe 840B or all-mpnet-base-v2⁸. We computed cosine

distance ($1 - \text{cosine similarity}$) instead of cosine similarity to match our other metrics. A higher mean pairwise sentence distance indicates that a text exhibits greater local variation of ideas within each text. Because this all-pair metric ignores sentence order, it should be interpreted as within-document semantic breadth or concentration rather than as a direct measure of narrative coherence, discourse structure, or writing quality.

Document-Level Distinctness. To assess the thematic distinctness of each individual text within a cohort, we adopted the minimum pairwise distance following Cox et al.⁶⁵ This metric quantifies how thematically unique a text is relative to other texts by identifying the smallest cosine distance between its document embedding and those of all others. We generated document embeddings using OpenAI’s *text-embedding-3-large* model⁷⁰ via the OpenAI API. This model was selected due to its high token limit (8,191 tokens), which accommodates the full length of college essays in our dataset. By contrast, other document embedding models used in earlier studies, such as *Universal Sentence Encoder* or SBERT’s *paraphrase-MiniLM-L6-v2*⁶⁶, support more limited input lengths. A higher minimum pairwise distance score indicates greater global distinctness of ideas in an entire text from other texts in a corpus. Because this is a cohort-relative measure of distinctness, it should not be interpreted as a comparison against an external historical baseline.

Semantic Level	Metric	Notation	Definition & Interpretation
Word-Level Diversity	DSI ³²	$\frac{1}{\binom{n_k}{2}} \sum_{i < j} d(w_{k,i}, w_{k,j})$	Computing the mean pairwise cosine distances between word embeddings within a text, using word-level BERT embeddings (specifically layers 6 and 7 of <i>bert-large-uncased</i>). Higher DSI scores indicate greater word-level semantic diversity.
Sentence-Level Diversity	Mean Pairwise Distance ⁸	$\frac{1}{\binom{m_k}{2}} \sum_{i < j} d(s_{k,i}, s_{k,j})$	Computing the mean pairwise cosine distances between sentence embeddings within each text, using sentence embeddings from the SBERT framework (<i>all-MiniLM-L6-v2 model</i>). A higher mean pairwise distance indicates greater local variation of ideas within each text.
Document-Level Distinctness	Minimum Pairwise Distance ^{65,66}	$\min_{j \in C_{t_k}^{(-k)}} d(v_k, v_j)$	Computing the smallest cosine distance between a text's document embedding and those of all the other texts in the same cohort, using OpenAI's <i>text-embedding-3-large</i> model. A higher minimum pairwise distance indicates a greater distance to the nearest counterpart, suggesting higher global distinctness of ideas in an entire text from other texts in a corpus.

Table 3.

Semantic Diversity Metrics: Definitions and Interpretations. n_k = number of word embeddings in

text k ; m_k = number of sentences in text k ; $w_{k,i}$, $s_{k,i}$ = word and sentence embeddings in text k ;

5 v_k = document embedding of text k ; $d(x, y)$ = cosine distance; $C_{t_k}^{(-k)}$ = set of texts in the same cohort as text k , excluding text k itself.

Natural Experiments

We employed a natural experiment design to assess the impact of generative AI on college admissions essays. We leveraged the public release of ChatGPT on November 30, 2022, as a temporal cutoff to define two distinct periods. Essays submitted in application cycles before this date (2018–2022 matriculation years) were classified as "pre-ChatGPT," while essays from cycles after this date were "post-ChatGPT" (2024–2025). We excluded all essays from the 2023 application cycle from our main analysis, except for descriptive statistics and robustness checks, because this application window overlapped with the release date, making it difficult to unambiguously classify those submissions as either pre- or post-ChatGPT¹.

College Application Datasets. We used comprehensive college admissions datasets containing all available personal statement essays and administrative records from applications to four U.S. universities: (1) an elite private university on the East Coast, (2) a regional private university on the West Coast, (3) a flagship public university in the Northeast, and (4) a selective liberal-arts college on the East Coast. As shown in Supplementary Table 1b, data for the 2018 cycle were only available for the selective liberal-arts college, and data for the 2019 cycle were only available for the selective liberal-arts college and the elite private university. The selective liberal-arts college had no data for the 2022 and 2023 cycles, and the regional private university lacked data for the 2025 cycle.

Data Cleaning and Sample Selection. To enhance comparability and mitigate the influence of outliers, we restricted the sample to essays falling within the middle 95% of word counts (2.5th–97.5th percentiles) for each institution as preregistered. During the processing of

¹ Admissions cycle years correspond to the matriculation year (e.g., the 2023 cycle is for students beginning university in Fall 2023). This cycle's application deadlines typically fall in the preceding months, with Early Decision/Action deadlines around early November in 2022, and Regular Decision deadlines around early January in 2023. ChatGPT's release on November 30, 2022, occurred after early applicants had likely completed their essays but before the regular deadline, thus confounding any clean pre- or post-treatment classification for this cohort.

the elite and regional private universities, we identified a small subset of essays that misplaced or omitted punctuation ($n \leq 6$), which prevented reliable sentence segmentation. Consequently, we preregistered an additional filter for the remaining two institutions (the flagship public university and the selective liberal-arts college), restricting the sample to the middle 95% of both word and sentence counts.

To maintain the independence of our confirmatory datasets, we excluded the specific samples used in our exploratory pilot studies, consistent with our preregistered analysis plan. This resulted in the removal of 2,100 essays from the elite private university, 1,500 from the regional private, 1,800 from the flagship public, and 1,800 from the selective liberal-arts college. In addition, because our document-level distinctness metric is based on nearest-neighbor distances within each admissions cycle, variations in sample sizes would confound year-to-year comparisons. Specifically, larger cohorts can naturally yield smaller distances simply due to increased density. To ensure meaningful year-to-year comparisons, we used random sampling to keep the number of essays consistent across cycles within each university, setting it to the smallest available count (see Supplementary Tables 1a–b). Detailed information on the resulting analytic sample for each university, including (1) demographic composition (Supplementary Tables 2a–d), (2) summary statistics for text features (Supplementary Tables 3a–d), and (3) applicant prompt selections where available (Supplementary Tables 4a–b), is provided below.

Primary pre/post results are shown in Fig. 1 and Supplementary Table 7. To verify that these sampling restrictions did not bias our results, we also replicated the analyses using the full, unbalanced dataset ($N = 394,199$ across four universities). This fuller sample yielded virtually identical patterns to our primary results, alongside even more pronounced document-level effects (Extended Data Fig. 1 and Supplementary Table 5), confirming that the observed trends were not driven by the specific exclusion criteria used in the main analyses.

Robustness and Diagnostic Analyses

We conducted additional analyses to test alternative explanations for the primary Semantic Disjunction pattern. We used GPTZero classifications in a balanced sample of 2,000 anonymized essays to compare pre-ChatGPT essays, post-ChatGPT essays classified as human-written, and post-ChatGPT essays classified as AI or mixed (see *Diagnostic Analysis* in Supplementary Information; Extended Data Fig. 2 and Supplementary Table 8). We estimated event-time models using admissions-cycle indicators with 2023 as the reference cycle, where available, to assess whether the observed shifts reflected pre-existing trends (see *Event-Time Analysis* in Supplementary Information; Extended Data Fig. 3 and Supplementary Tables 9a–c). We estimated COVID-period models comparing pre-COVID/pre-ChatGPT and post-COVID/post-ChatGPT periods against the pandemic period (see *COVID-19 Pandemic Confound Analysis* in Supplementary Information; Supplementary Tables 10a–d). We re-estimated the primary models with available covariates including essay length and demographic variables (see *Models with Control Variables* in Supplementary Information; Supplementary Tables 11a–d). Finally, we computed collective word-level diversity across essays by repeatedly sampling essays within institution-year-period cells, selecting one sentence per essay, pooling word-level BERT embeddings, and estimating post-minus-pre contrasts across nested sample sizes (see *Collective Word-Level Diversity Across Essays* in Supplementary Information; Extended Data Fig. 4).

Racial and Linguistic Subgroup Analyses

To assess whether Semantic Disjunction generalized across demographic and linguistic groups, we estimated separate regression models for each semantic level including ChatGPT period, subgroup membership, and their interaction. We evaluated generalizability using subgroup-specific simple effects of the ChatGPT period with Holm correction for multiple

comparisons (see *Racial and Linguistic Subgroup Analyses* in Supplementary Information; Fig. 2). We evaluated variability across groups using interaction terms relative to the White American reference group, defining a stronger Semantic Disjunction pattern as greater word-level diversity accompanied by steeper declines in sentence-level diversity and document-level distinctness (Extended Data Fig. 5 and Supplementary Tables 12a–e).

Investigating Mechanisms

In Controlled Experiment 1, we tested whether LLM writing weakens the link between word-level surface diversity and deeper conceptual diversity by constructing a prompt-matched comparison of human-written and GPT-generated essays. We sampled pre-ChatGPT human essays from institutions with prompt metadata and generated a GPT-5-mini essay for each matched prompt, matching the target word count of the human reference essay. We compared associations among word-, sentence-, and document-level semantic metrics across human-written and GPT-generated essays using correlation analyses (see *Investigating Mechanisms* in Supplementary Information; Supplementary Table 13a). To test whether this pattern also appeared in real-world writing, we replicated the coupling analysis in the natural-experiment data using period-by-semantic-metric interaction models (see *Investigating Mechanisms* in Supplementary Information; Supplementary Tables 13b–c).

Resolving Paradoxical Effects

To examine why post-ChatGPT essays were rated as more creative despite lower conceptual diversity, we first tested whether creativity ratings were more strongly associated with word-level surface diversity than with sentence- or document-level conceptual diversity. We analyzed expert ratings of 370 human-written essays and fine-tuned GPT-4.1-mini to predict expert percentile ratings so that the same expert-calibrated outcome could be applied to the natural-experiment essays (see *Resolving Paradoxical Effects* in Supplementary Information;

Table 1). We then estimated multiple regression models predicting creativity ratings from word-, sentence-, and document-level semantic metrics simultaneously, and used linear hypothesis tests to compare standardized coefficients (Table 2).

To assess the previously suggested possibility that higher creativity ratings for homogenized content result from evaluation of texts one at a time or in small groups (making homogenization less visible), Controlled Experiment 2 tested whether adding comparative context changed the association between semantic diversity levels and creativity ratings. The evaluator rated held-out essays after receiving 0 to 128 unlabeled reference essays, and we re-estimated the semantic-predictor models at each reference-set size (see *Resolving Paradoxical Effects* in Supplementary Information; Extended Data Fig. 6). Finally, to test whether post-ChatGPT changes in semantic diversity explained the increase in creativity ratings, we estimated parallel mediation models with word-, sentence-, and document-level diversity as simultaneous mediators (Fig. 3 and Supplementary Table 14).

Testing Semantic Disjunction Within-Subjects

In Controlled Experiment 3, we tested the causal effect of AI use on semantic diversity in a within-subjects design. We recruited 224 participants from the elite private university and retained 203 after exclusions for incomplete or invalid responses and failed attention checks. Participants uploaded their original admissions essays and wrote prompts for two AI-assisted conditions: revising their original essay and generating a new customized admissions essay. We generated AI-revised and AI-generated essays using GPT-4.1, then compared each participant's original, AI-revised, and AI-generated essays using mixed-effects models with participant random intercepts and Holm-corrected pairwise comparisons (see *Controlled Experiment 3* in Supplementary Information; Fig. 4 and Supplementary Tables 15a–b).

Acknowledgments

During the preparation of this work, the authors used generative AIs to generate GPT-assisted essays and improve the language and readability of the manuscript. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Funding: This work was supported by

Templeton World Charity Foundation TWCF-2023-32464 (KM, AEG, ALD)

National Science Foundation DRL-2400781 (KM, AEG, IV)

Author contributions:

Conceptualization: KM, AEG, KK, ALD

Methodology: KM, BL

Investigation: KM, AB, AEG, IV, JCK, DRJ

Visualization: KM

Funding acquisition: AEG, ALD

Project administration: KM

Supervision: KK, AEG, ALD

Writing – original draft: KM, AEG, KK

Writing – review & editing: KM, AEG, KK, BL, IV, JCK, DRJ, ALD

Competing Interests

The authors declare no competing interests.

Data and Materials Availability

The analyses were preregistered at the Open Science Framework (OSF; <https://osf.io/yd94z/registrations>) for natural experiments. While participant confidentiality constraints prevent the sharing of raw essay text, all processed data, code, and materials to reproduce the results of this study are available at <https://doi.org/10.17605/OSF.IO/YD94Z>.

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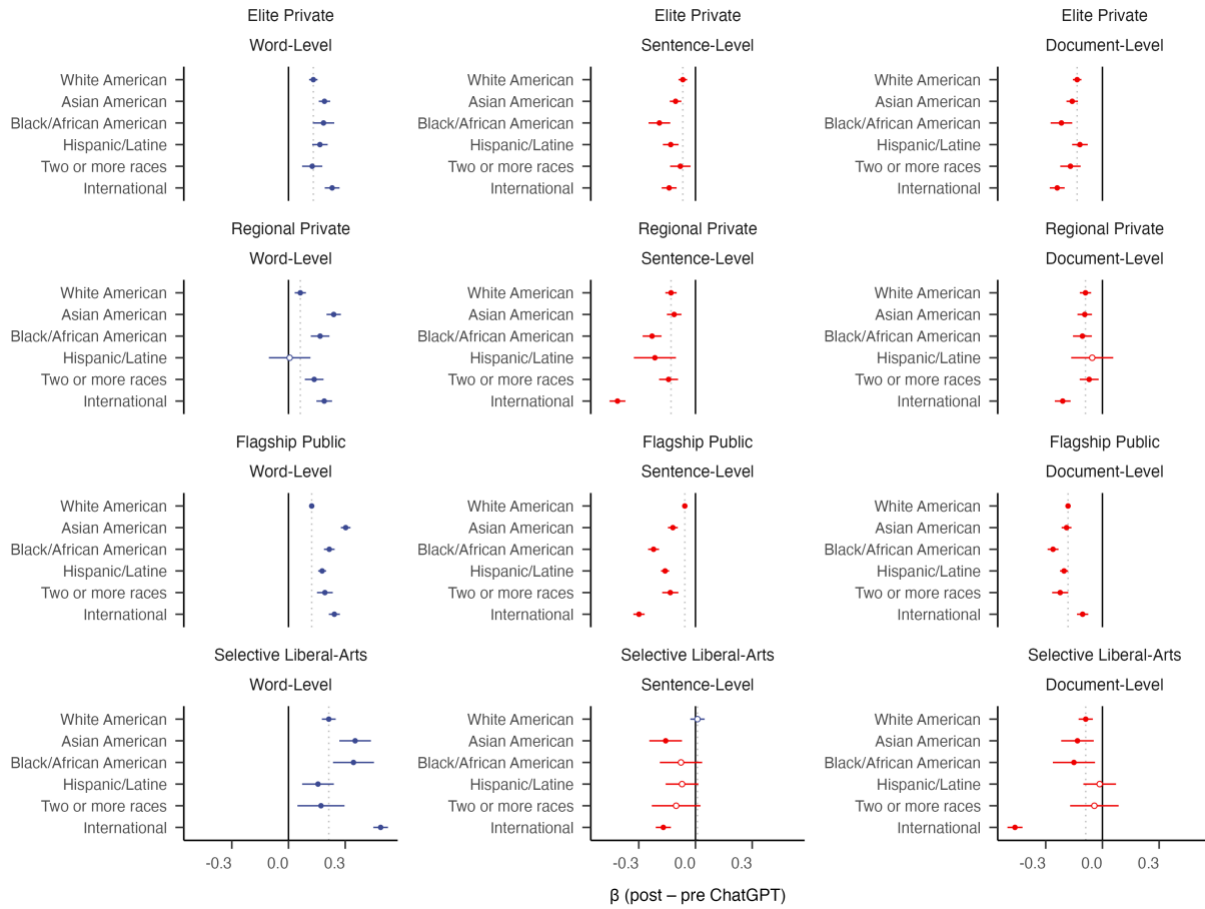
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Extended Data



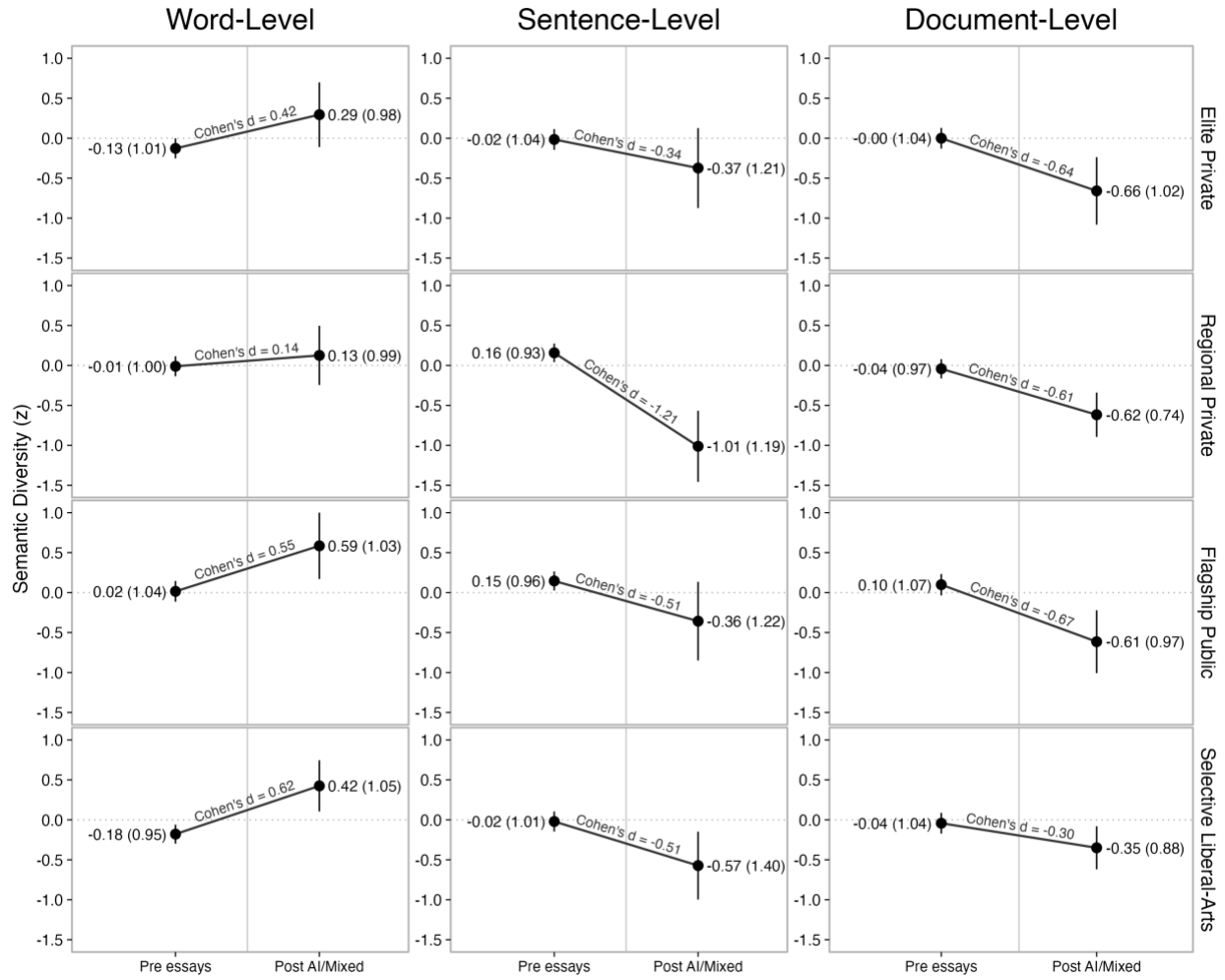
Extended Data Fig. 1.

Natural Experiment: Generalizability of Semantic Disjunction Across Racial and Ethnic

5 Subgroups Using Full Samples. $N = 394,199$ across four universities without per-cycle balancing. Forest plots display the estimated change in semantic diversity (β) following the release of ChatGPT (Post-ChatGPT minus Pre-ChatGPT) across four U.S. universities. Dots represent point estimates and error bars indicate 95% confidence intervals. Solid dots indicate statistically significant effects, after Holm correction for multiple comparisons within each semantic level. Hollow dots indicate non-significant effects. Grey dotted lines indicate the effect of the White American applicants within each university and semantic level. Applicants with

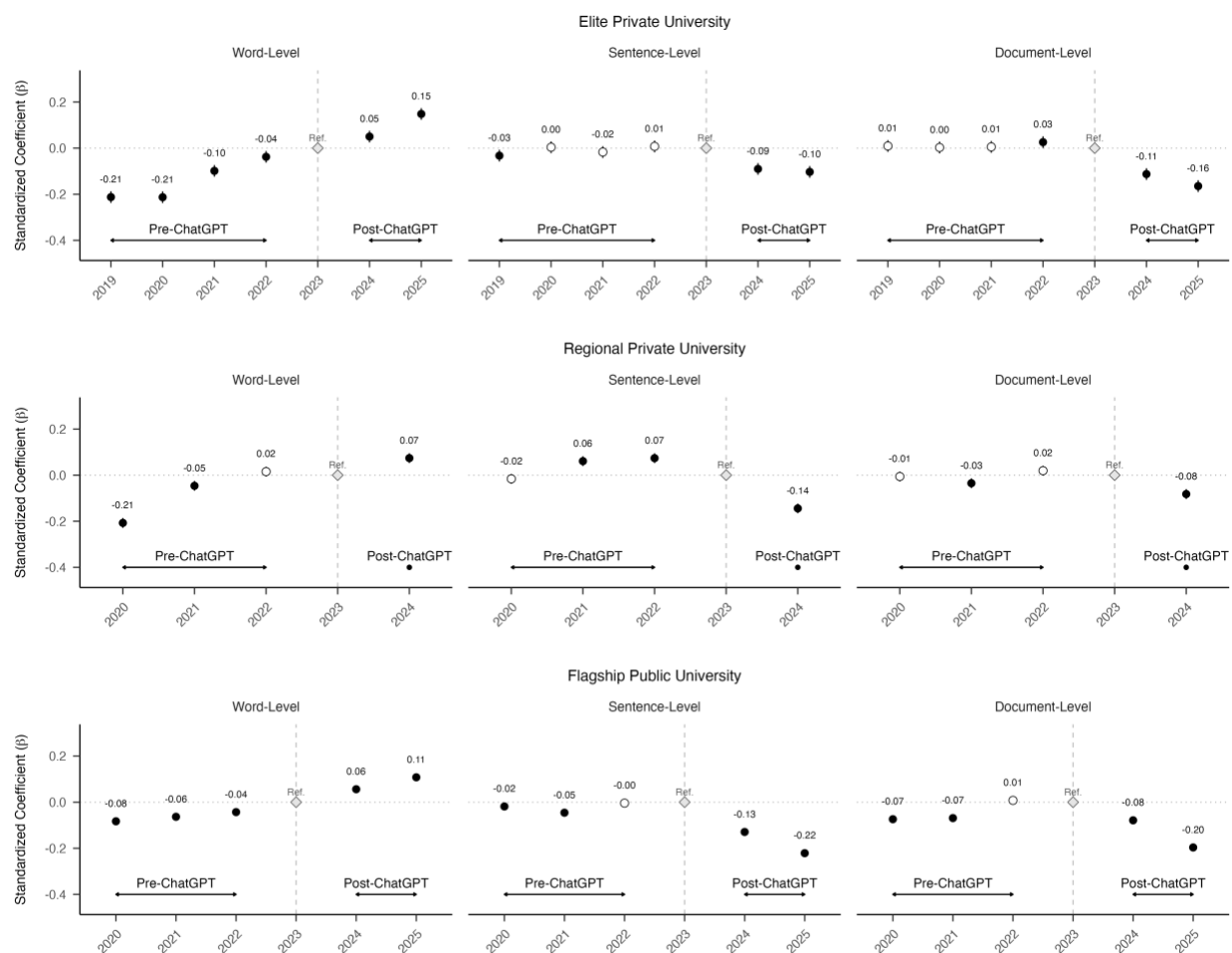
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Indigenous backgrounds or with missing/undisclosed information were included in the models but were omitted from the plots due to small subsample sizes.



Extended Data Fig. 2.

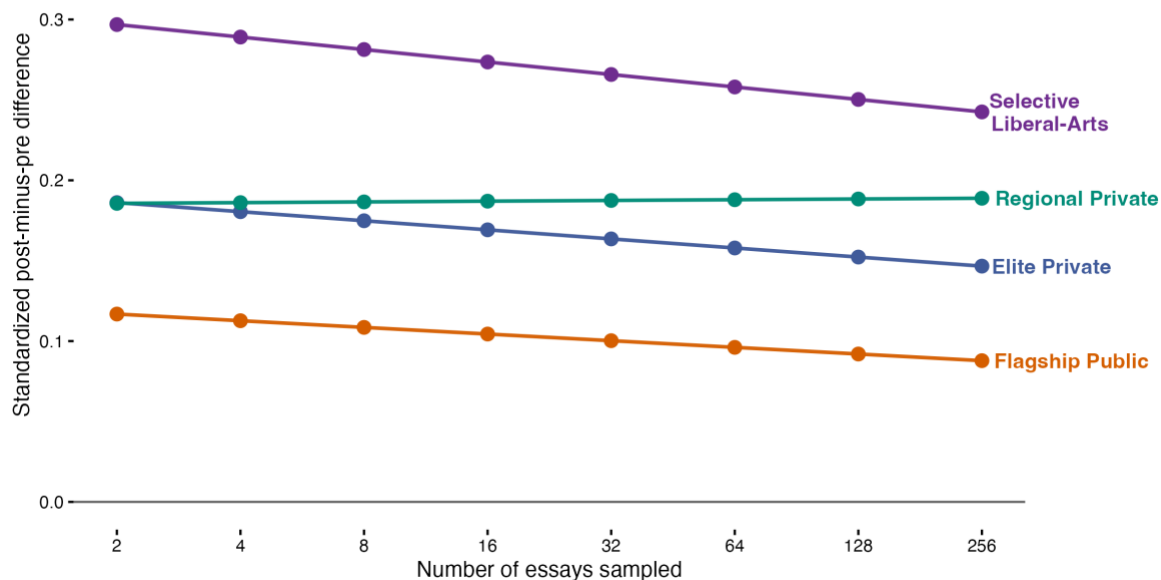
Natural Experiment: Semantic Disjunction in GPTZero-flagged post-ChatGPT essays. Points and error bars show mean Semantic Diversity (z) scores and 95% confidence intervals. Semantic diversity scores were standardized within each institution and semantic level. Pre-ChatGPT essays include 250 essays per institution. Post-ChatGPT AI/mixed essays are the subset classified by GPTZero as AI or mixed: elite private, $n = 25$; regional private, $n = 30$; flagship public, $n = 26$; selective liberal-arts, $n = 44$.



Extended Data Fig. 3.

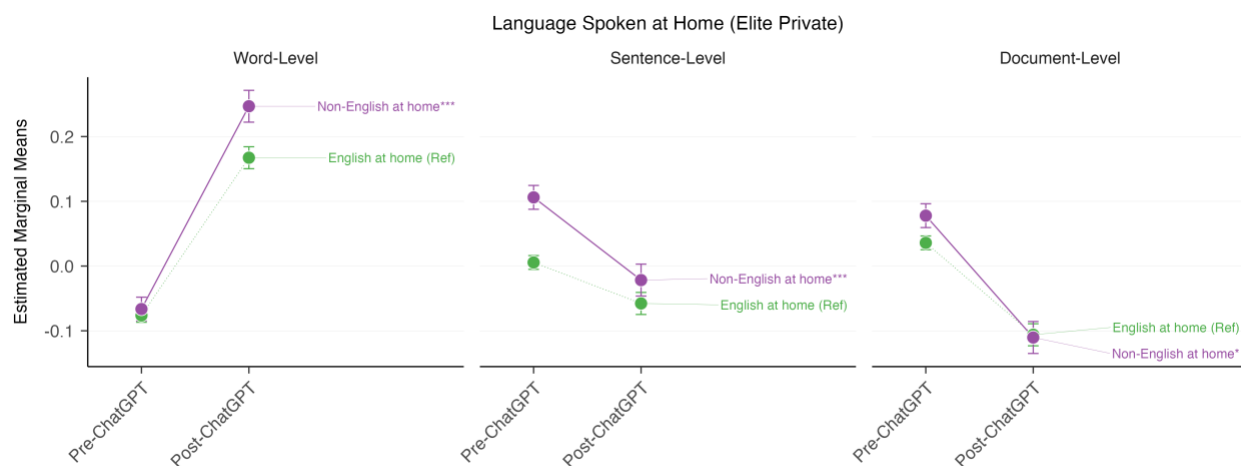
Natural Experiment: Event-Time Analyses of Semantic Diversity Pre- and Post-ChatGPT

Periods. Circles and error bars show the standardized coefficient for a given admissions cycle relative to the 2023 reference cycle (dotted line), with 95% confidence intervals, for the elite private university (top panel), the regional private university (middle panel), and the flagship public university (bottom panel). The 2023 reference cycle is marked as “Ref.” at $\beta = 0$. No event-time analysis was conducted for the selective liberal-arts college due to the absence of 2023 data. Semantic diversity scores were standardized within each institution and embedding granularity, using pooled data across all cycles. Filled circles indicate statistically significant differences from 2023 ($p < .05$); open circles indicate nonsignificant differences.



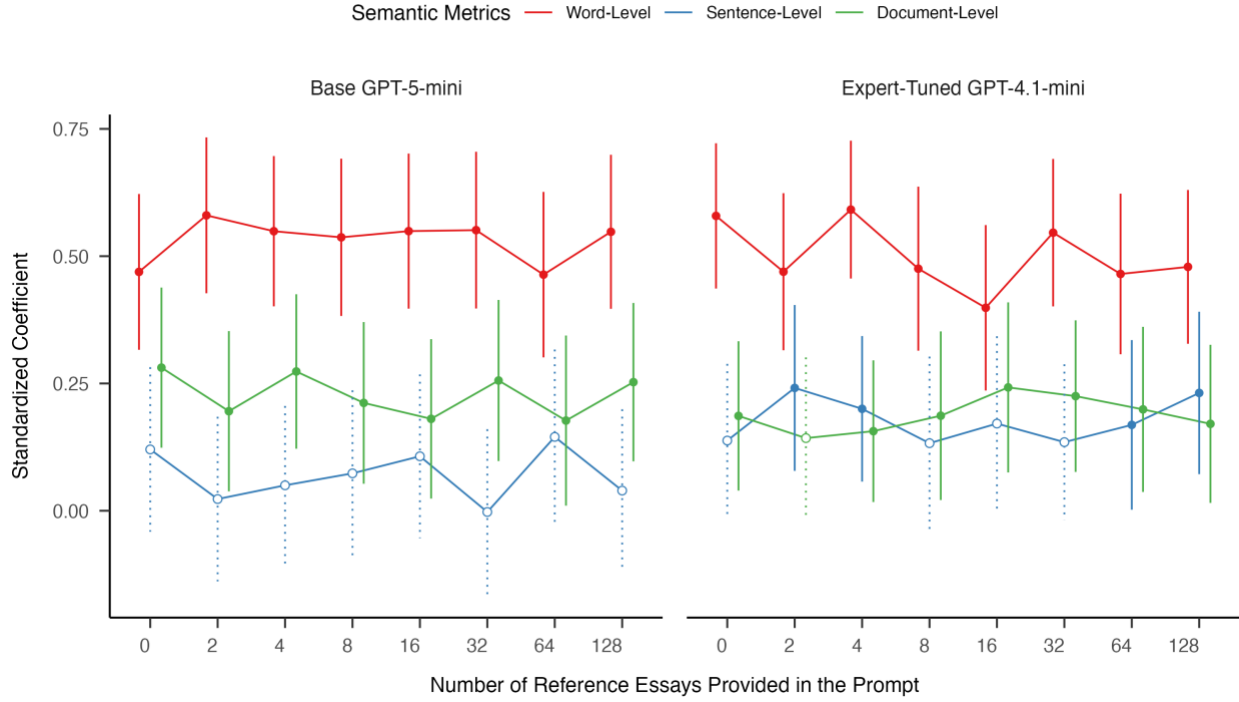
Extended Data Fig. 4.

Natural Experiment: Post-minus-pre differences in collective word-level diversity across essays. Points show institution-specific estimated post-minus-pre differences in collective word-level diversity across essays. Estimates were obtained from separate institution-specific models predicting standardized collective word-level diversity from ChatGPT period, $\log_2(N \text{ essays})$, and their interaction. From these models, we computed model-implied post-minus-pre contrasts at each essay sample size. Post-ChatGPT cohorts showed higher collective word-level diversity than pre-ChatGPT cohorts in every institution across all essay sample sizes tested, from 2 to 256 essays. All 32 model-estimated institution-by-sample-size contrasts were positive and statistically significant (all $ps < .027$), with standardized post-minus-pre differences ranging from 0.088 to 0.297.



Extended Data Fig. 5.

Natural Experiments: Semantic Diversity by Linguistic Backgrounds in Pre- and Post-ChatGPT Essays. Dots and error bars indicate the estimated marginal means and 95% confidence intervals of semantic diversity scores for each subgroup in the pre- and post-ChatGPT periods, measured separately at three semantic levels. Solid lines indicate significant differences in change compared with applicants from English-speaking households. See Supplementary Table 12e for full results. $*p < .05$. $***p < .001$.



Extended Data Fig. 6.

Controlled Experiment 2: Relationship Between Creativity Ratings and Semantic Metrics Across Varying In-Context Example Counts. $n = 111$. Standardized coefficients (β) derived from multiple linear regression models predicting creativity ratings of LLM evaluators from three semantic diversity metrics (word, sentence, and document levels). A separate regression model was fitted for each experimental condition in *Controlled Experiment 2*, defined by the number of reference essays provided in the prompt (ranging from 0 to 128), to determine the unique contribution of each semantic level to the model's creativity judgment in that specific context. Error bars represent 95% confidence intervals. Solid lines and filled points indicate statistically significant coefficients ($p < .05$), while dotted lines and hollow points indicate non-significant coefficients.

Analysis Phase	Hypothesis Description	Analysis Type	Dataset	Empirical Support
Semantic Disjunction	Post-ChatGPT essays show higher word-level diversity than pre-ChatGPT essays.	Pre	Natural Experiments EP, RP, FP, SL	Supported across all universities
	Post-ChatGPT essays show lower sentence-level diversity than pre-ChatGPT essays.	Pre	EP, RP, FP, SL	Supported across all universities
	Post-ChatGPT essays show lower document-level distinctness than pre-ChatGPT essays.	Pre	EP, RP, FP, SL	Supported across all universities
	Semantic disjunction generalizes across all subgroups.	Exp/Pre	EP = Exp; RP, FP, SL = Pre	Supported (EP, RP, FP), Partially Supported (SL)
	Semantic disjunction is stronger among non-mainstream groups.	Exp/Pre	EP = Exp; RP, FP, SL = Pre	Supported (EP, RP, FP), Partially Supported (SL)
Mechanisms & Evaluation	Decoupling: AI text reverses the positive human association of word-to-idea diversity.	Exp	Ctrl. Experiment 1: Matched Comparison (Human vs. GPT)	Supported
	Decoupling: Post-ChatGPT essays show weaker associations of word-to-idea diversity than pre-ChatGPT essays.	Exp	EP, RP, FP, SL	Supported (RP, FP), Partially Supported (EP, SL)
	Lexical Bias: Evaluators rely disproportionately on surface-level lexical cues.	Exp	Assessment Analysis Human Experts, LLM Evaluators	Supported
	Increased word-level diversity accounts for increased creativity ratings for post-ChatGPT essays, despite conceptual homogenization at the sentence and document levels.	Exp/Pre	EP = Exp; RP, FP, SL = Pre	Supported across all universities
	Joint Evaluation: Providing comparative context (0 to 128 reference essays) does not mitigate lexical bias.	Exp	Ctrl. Experiment 2: LLM Evaluators	Supported
Causal Evidence	AI use (revision/generation) causally produces semantic disjunction.	Exp	Ctrl. Experiment 3: Controlled Within-Subjects Experiment	Supported
	AI-revision preserves more distinctness than AI-generation.	Exp		Supported

Extended Data Table 1.

Research Hypotheses and Outcomes Across Universities. Pre = Preregistered; Exp =

Exploratory; EP = Elite Private University; RP = Regional Private University; FP = Flagship

Public University; SL = Selective Liberal-Arts College. Ctrl. Experiment = Controlled

5 Experiment. LLM Evaluators included GPT-5-mini and Expert-Tuned GPT-4.1-mini. Full preregistration documents for all four institutions are available on the Open Science Framework at <https://osf.io/yd94z>.

Supplementary Information for

The Link Between Diverse Words and Original Ideas Is Weakening in AI-Era College Admissions

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Supplementary Information

Sample Characteristics, Descriptives, and Pilot Study

Institution	Exploratory pilot essays removed	Eligible full sample	Balanced n retained per cycle	Removed by per-cycle balancing	Balanced sample retained	2023 transition-cycle essays	Primary pre/post analytic N
Elite Private	2,100	101,529	11,880	18,369	83,160	11,880	71,280
Regional Private	1,500	89,712	16,595	6,737	82,975	16,595	66,380
Flagship Public	1,800	249,441	30,930	63,861	185,580	30,930	154,650
Selective Liberal-Arts	1,800	29,885	3,513	8,807	21,078	0	21,078
Total	7,200	470,567		97,774	372,793	59,405	313,388

Supplementary Table 1a.

Natural Experiments: Sample selection flow by institution. “Eligible full sample” refers to the unbalanced sample available after preregistered cleaning filters and removal of essays previously used in exploratory pilot analyses. “Removed by per-cycle balancing” refers only to random down-sampling used to equalize the number of essays across application cycles within each institution. The 2023 transition cycle was processed and retained in the balanced sample, where available, but excluded from the primary pre/post-ChatGPT analyses as preregistered.

Institution	Balanced <i>n</i> retained per cycle	2018	2019	2020	2021	2022	2023	2024	2025
Elite Private	11,880	-	11,880 / 13,069	11,880 / 11,880	11,880 / 16,486	11,880 / 15,123	11,880 / 14,427	11,880 / 15,009	11,880 / 15,535
Regional Private	16,595	-	-	16,595 / 17,242	16,595 / 16,595	16,595 / 17,563	16,595 / 19,933	16,595 / 18,379	-
Flagship Public	30,930	-	-	30,930 / 30,930	30,930 / 33,577	30,930 / 37,350	30,930 / 42,008	30,930 / 50,112	30,930 / 55,464
Selective Liberal-Arts	3,513	3,513 / 4,501	3,513 / 4,503	3,513 / 3,513	3,513 / 4,830	-	-	3,513 / 5,853	3,513 / 6,685

Supplementary Table 1b.

Natural Experiments: Year-by-year per-cycle balancing. Cells report retained essays / eligible essays after preregistered cleaning filters and removal of essays previously used in exploratory pilot analyses. Cells with no additional exclusions, such as 11,880 / 11,880, indicate that the institution-year was the smallest eligible cycle and therefore set the balanced per-cycle sample size; they do not imply that no essays were removed during earlier cleaning or pilot-sample exclusion stages.

	2019	2020	2021	2022	2023	2024	2025
Race							
Asian American	1,954 (16%)	2,151 (18%)	2,081 (18%)	2,510 (21%)	2,629 (22%)	2,583 (22%)	2,560 (22%)
Black/African American	732 (6%)	818 (7%)	695 (6%)	705 (6%)	733 (6%)	678 (6%)	629 (5%)
Hispanic/Latine	1,428 (12%)	1,436 (12%)	1,326 (11%)	1,270 (11%)	1,296 (11%)	1,251 (11%)	1,324 (11%)
Indigenous Peoples	23 (0%)	21 (0%)	18 (0%)	19 (0%)	24 (0%)	29 (0%)	20 (0%)
International	1,167 (10%)	1,233 (10%)	1,563 (13%)	1,344 (11%)	1,304 (11%)	1,461 (12%)	1,586 (13%)
White American	5,447 (46%)	5,154 (43%)	5,155 (43%)	4,864 (41%)	4,674 (39%)	4,169 (35%)	3,933 (33%)
Two or more races	762 (6%)	723 (6%)	731 (6%)	765 (6%)	766 (6%)	774 (7%)	754 (6%)
Prefer not to disclose	244 (2%)	291 (2%)	310 (3%)	400 (3%)	454 (4%)	933 (8%)	1,067 (9%)
Missing	123 (1%)	53 (0%)	1 (0%)	3 (0%)	0 (0%)	2 (0%)	7 (0%)
Sex							
Female	7,245 (61%)	7,186 (60%)	7,247 (61%)	7,135 (60%)	7,187 (60%)	6,834 (58%)	6,867 (58%)
Male	4,613 (39%)	4,657 (39%)	4,586 (39%)	4,644 (39%)	4,552 (38%)	4,939 (42%)	4,926 (41%)
Non-binary	9 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Missing	13 (0%)	37 (0%)	47 (0%)	101 (1%)	141 (1%)	107 (1%)	87 (1%)
English Spoken at Home							
	8,786 (76%)	8,612 (75%)	8,530 (75%)	7,934 (74%)	6,816 (70%)	6,671 (68%)	6,485 (67%)
Missing	363 (3%)	384 (3%)	556 (5%)	1,092 (9%)	2,197 (18%)	2,119 (18%)	2,216 (19%)

Supplementary Table 2a.

Natural Experiment: Demographic Composition of the Final Analytic Sample (Elite Private University). $N = 71,280$ (11,880 applications per admissions cycle): 47,520 essays from 2019–2022 (pre-ChatGPT) and 23,760 from 2024–2025 (post-ChatGPT). Data from the 2023 transitional cycle ($n = 11,880$) were processed but excluded from the primary analysis to ensure temporal distinction. Race and ethnicity were not available for applicants who were not U.S. citizens.

	2020	2021	2022	2023	2024
Race					
Asian American	3,101 (19%)	3,109 (19%)	3,090 (19%)	3,041 (18%)	3,117 (19%)
Black/African American	1,618 (10%)	1,730 (10%)	1,790 (11%)	1,706 (10%)	1,909 (12%)
Hispanic/Latine	397 (2%)	541 (3%)	332 (2%)	338 (2%)	363 (2%)
Indigenous Peoples	235 (1%)	291 (2%)	249 (2%)	283 (2%)	264 (2%)
International	2,504 (15%)	2,350 (14%)	2,381 (14%)	2,768 (17%)	2,737 (16%)
White American	5,346 (32%)	5,376 (32%)	5,620 (34%)	5,475 (33%)	5,018 (30%)
Two or more races	1,985 (12%)	1,764 (11%)	1,825 (11%)	1,901 (11%)	1,832 (11%)
Missing	1,409 (8%)	1,434 (9%)	1,308 (8%)	1,083 (7%)	1,355 (8%)
Sex					
Female	10,773 (65%)	11,081 (67%)	11,030 (66%)	10,802 (65%)	10,920 (66%)
Male	5,822 (35%)	5,514 (33%)	5,565 (34%)	5,793 (35%)	5,674 (34%)
Missing	0 (0%)	0 (0%)	0 (0%)	0 (0%)	1 (0%)

Supplementary Table 2b.

Natural Experiment: Demographic Composition of the Final Analytic Sample (Regional Private University). $N = 66,380$ (16,595 applications per admissions cycle): 49,785 essays from 2020–2022 (pre-ChatGPT) and 16,595 from 2024 (post-ChatGPT). Data from the 2023 transitional cycle ($n = 16,595$) were processed but excluded from the primary analysis to ensure temporal distinction. Race and ethnicity were reported for applicants who were not U.S. citizens but were recoded as *International* to ensure consistency with the institution’s standard classification. Data were not available for the 2018 and 2019 cycles.

	2020	2021	2022	2023	2024	2025
Race						
Asian American	3,129 (10%)	3,176 (10%)	3,286 (11%)	3,549 (11%)	3,439 (11%)	3,580 (12%)
Black/African American	2,387 (8%)	2,498 (8%)	2,610 (8%)	2,634 (9%)	2,861 (9%)	3,089 (10%)
Hispanic/Latine	4,179 (14%)	4,211 (14%)	4,479 (14%)	4,854 (16%)	5,486 (18%)	5,781 (19%)
Indigenous Peoples	43 (0%)	38 (0%)	44 (0%)	56 (0%)	57 (0%)	70 (0%)
International	3,981 (13%)	2,830 (9%)	2,602 (8%)	2,408 (8%)	2,282 (7%)	1,993 (6%)
White American	15,370 (50%)	16,090 (52%)	15,969 (52%)	15,857 (51%)	14,808 (48%)	14,137 (46%)
Two or more races	1,052 (3%)	1,146 (4%)	1,125 (4%)	1,209 (4%)	1,419 (5%)	1,500 (5%)
Missing	789 (3%)	941 (3%)	815 (3%)	363 (1%)	578 (2%)	780 (3%)
Sex						
Female	16,786 (54%)	17,181 (56%)	17,472 (56%)	18,004 (58%)	18,045 (58%)	18,327 (59%)
Male	13,752 (44%)	13,297 (43%)	13,116 (42%)	12,925 (42%)	12,772 (41%)	12,507 (40%)
Missing	392 (1%)	452 (1%)	342 (1%)	1 (0%)	113 (0%)	96 (0%)

Supplementary Table 2c.

Natural Experiment: Demographic Composition of the Final Analytic Sample (Flagship Public University). $N = 154,650$ (30,930 applications per admissions cycle): 92,790 essays from 2020–2022 (pre-ChatGPT) and 61,860 from 2024–2025 (post-ChatGPT). Data from the 2023 transitional cycle ($n = 30,930$) were processed but excluded from the primary analysis to ensure temporal distinction. Race and ethnicity were not available for applicants who were not U.S. citizens. Data were not available for the 2018 and 2019 cycles.

	2018	2019	2020	2021	2024	2025
Race						
Asian American	347 (10%)	336 (10%)	262 (7%)	314 (9%)	263 (7%)	175 (5%)
Black/African American	230 (7%)	189 (5%)	154 (4%)	197 (6%)	127 (4%)	123 (4%)
Hispanic/Latine	287 (8%)	331 (9%)	260 (7%)	254 (7%)	243 (7%)	214 (6%)
Indigenous Peoples	11 (0%)	7 (0%)	1 (0%)	11 (0%)	3 (0%)	2 (0%)
International	581 (17%)	695 (20%)	892 (25%)	954 (27%)	1,567 (45%)	1,844 (52%)
White American	1,847 (53%)	1,774 (50%)	1,768 (50%)	1,586 (45%)	1,123 (32%)	1,013 (29%)
Two or more races	141 (4%)	130 (4%)	120 (3%)	137 (4%)	105 (3%)	98 (3%)
Missing	69 (2%)	51 (1%)	56 (2%)	60 (2%)	82 (2%)	44 (1%)
Sex						
Female	N/A	N/A	N/A	N/A	1,709 (49%)	1,669 (48%)
Male	N/A	N/A	N/A	N/A	1,804 (51%)	1,844 (52%)

Supplementary Table 2d.

Natural Experiment: Demographic Composition of the Final Analytic Sample (Selective Liberal-Arts College). $N = 21,078$ (3,513 applications per admissions cycle): 14,052 essays from 2018–2021 (pre-ChatGPT) and 7,026 from 2024–2025 (post-ChatGPT). The application datasets for the 2022 and 2023 cycles were not available at this institution. Race and ethnicity were not reported for applicants who were not U.S. citizens. Sex was not available in the 2018–2021 cycles.

	2019	2020	2021	2022	2023	2024	2025
Word Count							
Mean (SD)	619 (58)	619 (58)	625 (54)	623 (54)	625 (52)	622 (56)	622 (54)
Range	432, 740	432, 740	432, 740	432, 740	432, 740	432, 740	432, 740
Sentence Count							
Mean (SD)	33 (8)	34 (8)	34 (8)	34 (8)	34 (8)	34 (8)	34 (8)
Range	2, 82	1, 86	1, 80	1, 91	1, 81	1, 86	3, 91
Word-Level							
Mean (SD)	0.830 (0.008)	0.830 (0.008)	0.831 (0.008)	0.831 (0.008)	0.832 (0.008)	0.832 (0.008)	0.833 (0.008)
Range	0.791, 0.861	0.801, 0.856	0.792, 0.858	0.794, 0.861	0.790, 0.856	0.796, 0.858	0.784, 0.859
Sentence-Level							
Mean (SD)	0.77 (0.05)	0.77 (0.05)	0.77 (0.04)	0.77 (0.05)	0.77 (0.04)	0.77 (0.05)	0.77 (0.05)
Range	0.54, 0.89	0.42, 0.90	0.22, 0.89	0.27, 0.89	0.21, 0.89	0.41, 0.89	0.23, 0.89
Document-Level							
Mean (SD)	0.23 (0.05)	0.23 (0.05)	0.23 (0.05)	0.23 (0.05)	0.23 (0.05)	0.22 (0.05)	0.22 (0.05)
Range	0.06, 0.52	0.08, 0.72	0.05, 0.50	0.09, 0.52	0.08, 0.74	0.07, 0.74	0.08, 0.56

Supplementary Table 3a.

Natural Experiment: Summary Statistics for Text Features and Semantic Metrics Across Years (Elite Private University). $N = 83,160$ (11,880 applications per admissions cycle). All applicants responded to the same prompt: “*As OOOO University is a diverse community, the Admissions Committee would like to know more about you in your words. Please submit a brief essay, either personal or creative, which you feel best describes you.*” The average essay length was 622.03 words ($SD = 55.23$), with significant but negligible variation by admissions cycle, $F(6, 83153) = 22.21, p < .001, \eta^2_g = .002$.

	2020	2021	2022	2023	2024
Word Count					
Mean (SD)	575 (95)	605 (113)	601 (111)	583 (92)	584 (90)
Range	291, 855	291, 856	291, 856	291, 818	291, 654
Sentence Count					
Mean (SD)	31 (9)	32 (9)	32 (9)	31 (9)	31 (9)
Range	2, 95	1, 101	1, 105	1, 119	1, 119
Word-Level					
Mean (SD)	0.822 (0.008)	0.824 (0.008)	0.824 (0.008)	0.824 (0.009)	0.825 (0.009)
Range	0.772, 0.861	0.759, 0.856	0.789, 0.854	0.774, 0.851	0.769, 0.856
Sentence-Level					
Mean (SD)	0.75 (0.05)	0.75 (0.05)	0.75 (0.05)	0.75 (0.05)	0.74 (0.06)
Range	0.39, 0.89	0.31, 0.89	0.40, 0.89	0.33, 0.89	0.00, 1.02
Missing	1	0	2	2	1
Document-Level					
Mean (SD)	0.21 (0.05)	0.21 (0.05)	0.21 (0.05)	0.21 (0.05)	0.21 (0.05)
Range	0.08, 0.61	0.06, 0.62	0.09, 0.62	0.00, 0.65	0.00, 0.67

Supplementary Table 3b.

Natural Experiment: Summary Statistics for Text Features and Semantic Metrics Across Years (Regional Private University). $N = 82,975$ (16,595 applications per admissions cycle). Six essays were missing sentence-level diversity scores because misplaced or omitted punctuation prevented sentence segmentation; word-level diversity and document-level distinctness were still computed for these cases. The average essay length was 589.38 words ($SD = 101.28$), with significant but small variation across admissions cycles, $F(4, 82970) = 268.26, p < .001, \eta^2g = .001$. In each cycle, applicants could choose one of the seven essay prompts, with the option to write on a topic of their own choice. The full distribution of prompt selection is provided in Supplementary Table 4a.

	2020	2021	2022	2023	2024	2025
Word Count						
Mean (SD)	606 (55)	611 (52)	608 (54)	610 (53)	610 (53)	612 (50)
Range	394, 650	394, 650	394, 650	394, 650	394, 650	394, 650
Sentence Count						
Mean (SD)	32 (7)	32 (7)	33 (7)	33 (7)	33 (7)	33 (7)
Range	18, 50	18, 50	18, 50	18, 50	18, 50	18, 50
Word-Level						
Mean (SD)	0.824 (0.008)	0.825 (0.008)	0.825 (0.008)	0.825 (0.008)	0.825 (0.008)	0.826 (0.008)
Range	0.793, 0.853	0.786, 0.857	0.774, 0.855	0.788, 0.856	0.792, 0.854	0.784, 0.869
Sentence-Level						
Mean (SD)	0.76 (0.05)	0.76 (0.05)	0.76 (0.05)	0.76 (0.05)	0.76 (0.05)	0.75 (0.05)
Range	0.45, 0.89	0.49, 0.89	0.47, 1.10	0.49, 0.89	0.38, 0.90	0.47, 0.89
Missing	1	0	0	0	0	0
Document-Level						
Mean (SD)	0.20 (0.05)	0.20 (0.05)	0.20 (0.05)	0.20 (0.05)	0.20 (0.05)	0.19 (0.05)
Range	0.07, 0.65	0.07, 0.55	0.07, 0.64	0.01, 0.60	0.05, 0.62	0.06, 0.54

Supplementary Table 3c.

Natural Experiment: Summary Statistics for Text Features and Semantic Metrics Across Years (Flagship Public University). $N = 185,580$ (30,930 applications per admissions cycle). One essay was missing a sentence-level diversity score; word-level diversity and document-level distinctness were still computed. The average essay length was 609.47 words ($SD = 52.84$), with significant but small variation across admissions cycles, $F(5, 185574) = 50.69, p < .001, \eta^2_g = .013$. Common App essay prompts were used, but the specific prompt choice associated with each essay was unavailable for this university.

	2018	2019	2020	2021	2024	2025
Word Count						
Mean (SD)	601 (64)	609 (58)	611 (54)	614 (53)	615 (51)	620 (46)
Range	368, 650	372, 650	373, 650	369, 650	368, 650	368, 650
Sentence Count						
Mean (SD)	33 (7)	33 (7)	32 (7)	32 (7)	32 (7)	33 (7)
Range	17, 50	17, 50	17, 50	17, 50	17, 50	17, 50
Word-Level						
Mean (SD)	0.827 (0.008)	0.827 (0.008)	0.828 (0.008)	0.828 (0.008)	0.830 (0.008)	0.831 (0.008)
Range	0.797, 0.856	0.799, 0.851	0.797, 0.852	0.795, 0.860	0.797, 0.855	0.797, 0.856
Sentence-Level						
Mean (SD)	0.77 (0.05)	0.77 (0.05)	0.77 (0.05)	0.77 (0.05)	0.76 (0.05)	0.77 (0.05)
Range	0.53, 0.88	0.55, 0.89	0.50, 0.88	0.52, 0.89	0.47, 0.88	0.48, 0.89
Document-Level						
Mean (SD)	0.25 (0.06)	0.25 (0.06)	0.25 (0.05)	0.25 (0.06)	0.25 (0.06)	0.24 (0.05)
Range	0.10, 0.56	0.09, 0.69	0.10, 0.54	0.10, 0.53	0.06, 0.54	0.10, 0.55

Supplementary Table 3d.

Natural Experiment: Summary Statistics for Text Features and Semantic Metrics Across Years (Selective Liberal-Arts College). $N = 21,078$ (3,513 applications per admissions cycle). The application datasets for the 2022–2023 cycles were not available at this institution. The average essay length was 611.62 words ($SD = 54.72$), with significant but small variation across admissions cycles, $F(5, 21072) = 46.16, p < .001, \eta^2g = .011$. In each cycle, applicants could choose one of the seven essay prompts, with the option to write on a topic of their own choice. The full distribution of prompt selection is provided in Supplementary Table 4b.

	2020	2021	2022	2023	2024
Applicant Choice	3,546 (21%)	3,559 (21%)	4,118 (25%)	3,978 (24%)	4,046 (24%)
Challenged Belief Prompt	525 (3%)	555 (3%)	556 (3%)	530 (3%)	564 (3%)
Childhood Adulthood Transition	3,648 (22%)	3,747 (23%)	3,367 (20%)	3,432 (21%)	3,401 (20%)
Content Environment Experience	630 (4%)	596 (4%)	0 (0%)	0 (0%)	0 (0%)
Engaging Lost Time	692 (4%)	812 (5%)	936 (6%)	906 (5%)	856 (5%)
Failure Incident	3,753 (23%)	3,835 (23%)	3,735 (23%)	3,735 (23%)	3,597 (22%)
Reflect Gratitude	0 (0%)	2 (0%)	532 (3%)	547 (3%)	541 (3%)
Story Identity Central	3,801 (23%)	3,489 (21%)	3,323 (20%)	3,465 (21%)	3,574 (22%)
Missing	0 (0%)	0 (0%)	28 (0%)	2 (0%)	16 (0%)

Supplementary Table 4a.

Natural Experiment: Distribution of Essay Prompt Selection by Admissions Cycle (Regional Private University). $N = 82,975$ (16,595 applications per admissions cycle). Full prompt descriptions are available from the Common Application at <https://www.commonapp.org>.

	2018	2019	2020	2021	2024	2025
Applicant Choice	779 (22%)	833 (24%)	887 (25%)	900 (26%)	1,072 (31%)	1,061 (30%)
Challenged Belief Prompt	104 (3%)	91 (3%)	127 (4%)	107 (3%)	99 (3%)	108 (3%)
Childhood Adulthood Transition	905 (26%)	867 (25%)	844 (24%)	842 (24%)	762 (22%)	770 (22%)
Content Environment Experience	113 (3%)	142 (4%)	119 (3%)	126 (4%)	0 (0%)	0 (0%)
Engaging Lost Time	141 (4%)	138 (4%)	133 (4%)	131 (4%)	115 (3%)	103 (3%)
Failure Incident	618 (18%)	658 (19%)	617 (18%)	650 (19%)	617 (18%)	646 (18%)
Reflect Gratitude	0 (0%)	0 (0%)	0 (0%)	0 (0%)	92 (3%)	82 (2%)
Story Identity Central	853 (24%)	784 (22%)	786 (22%)	757 (22%)	756 (22%)	743 (21%)

Supplementary Table 4b.

Natural Experiment: Distribution of Essay Prompt Selection by Admissions Cycle (Selective Liberal-Arts College). $N = 21,078$ (3,513 applications per admissions cycle). Full prompt descriptions are available from the Common Application at <https://www.commonapp.org>.

University	Semantic Level	Post-ChatGPT (n)	Pre-ChatGPT (n)	Post-ChatGPT	Pre-ChatGPT	<i>t</i>	df	Cohen's <i>d</i> [95% CI]
Elite Private	Word	30,530	56,012	0.12 (0.85)	-0.07 (1.07)	28.95	75645.75	0.20 [0.18, 0.21]
	Sentence	30,543	56,555	-0.06 (1.02)	0.03 (0.99)	-11.93	60671.16	-0.09 [-0.10, -0.07]
	Document	30,544	56,558	-0.10 (0.97)	0.06 (1.01)	-23.02	64793.51	-0.16 [-0.18, -0.15]
Regional Private	Word	18,379	51,400	0.11 (1.01)	-0.04 (0.99)	17.27	31900.47	0.15 [0.13, 0.17]
	Sentence	18,376	51,397	-0.14 (1.06)	0.05 (0.97)	-20.76	30143.39	-0.18 [-0.20, -0.16]
	Document	18,379	51,400	-0.07 (0.98)	0.03 (1.01)	-11.50	33231.70	-0.10 [-0.12, -0.08]
Flagship Public	Word	105,576	101,857	0.07 (1.00)	-0.08 (0.99)	34.74	207244.3	0.15 [0.14, 0.16]
	Sentence	105,576	101,856	-0.06 (1.02)	0.06 (0.97)	-27.95	207403.9	-0.12 [-0.13, -0.11]
	Document	105,576	101,857	-0.09 (0.94)	0.09 (1.05)	-39.77	203323.2	-0.18 [-0.18, -0.17]
Selective Liberal-Arts	Word	12,538	17,347	0.23 (1.00)	-0.17 (0.96)	34.39	26337.43	0.40 [0.38, 0.43]
	Sentence	12,538	17,347	-0.01 (1.02)	0.01 (0.98)	-1.67	26311.76	-0.02 [-0.04, 0.00]
	Document	12,538	17,347	-0.11 (0.95)	0.08 (1.03)	-16.07	28165.77	-0.19 [-0.21, -0.17]

Supplementary Table 5.

Natural Experiment: Semantic Diversity Changes in College Admissions Essays Using Full Samples (Pre- and Post-ChatGPT Differences). All Welch's *t*-tests compared standardized semantic diversity scores between pre- and post-ChatGPT essays. All contrasts were significant at $p < .001$ except sentence-level diversity at the selective liberal-arts college, $t(26311.76) = -1.67, p = .095$. The 2023 transition cycle was excluded from these analyses.

Pilot Study

We utilized a subset of 2,100 college admissions essays randomly drawn from archival submissions to an elite private university on the East Coast. Specifically, our pilot analysis included 1,800 essays: 1,200 essays from the 2019–2022 admissions cycles (pre-ChatGPT) and 600 from the 2024–2025 cycles (post-ChatGPT), with 300 essays randomly sampled per cycle. We sampled 300 essays from the 2023 cycle to compute descriptive statistics but excluded them as we did in our main analysis. For the selected essays, the mean length was 619.13 words ($SD = 72.38$), with no significant variation in length by admissions cycle, $F(6, 2093) = 1.61, p = .139$.

We observed diverging patterns in semantic diversity following the release of ChatGPT, depending on embedding granularities (see Supplementary Table 6). Specifically, post-ChatGPT essays exhibited significantly greater word-level diversity, $t(1199.468) = 5.19, p < .001$, Cohen's $d = 0.26$, 95% CI [0.16, 0.36], indicating increased lexical variation compared to pre-ChatGPT essays. In contrast, sentence-level diversity declined significantly, $t(1103.865) = -2.51, p = .012$, Cohen's $d = -0.13$, 95% CI [-0.23, -0.03], suggesting that ideas within each essay became more similar. Document-level distinctness also decreased significantly, $t(1259.365) = -4.93, p < .001$, Cohen's $d = -0.24$, 95% CI [-0.35, -0.14], indicating a reduction in thematic distinctness across essays.

Semantic Level	Post-ChatGPT	Pre-ChatGPT	<i>t</i>	df	<i>p</i>	Cohen's <i>d</i> [95% CI]
Word	0.17 (1.00)	-0.09 (1.00)	5.19	1199.468	< .001	0.26 [0.16, 0.36]
Sentence	-0.10 (1.07)	0.03 (0.97)	-2.51	1103.865	0.012	-0.13 [-0.23, -0.03]
Document	-0.17 (0.96)	0.07 (1.02)	-4.93	1259.365	< .001	-0.24 [-0.35, -0.14]

Supplementary Table 6.

Natural Experiment: Changes in Semantic Diversity in College Admissions Essays: Pre- and Post-ChatGPT Differences by Embedding Granularity (Pilot Study). Welch's *t*-test was used to compare the means of standardized semantic diversity between pre-ChatGPT essays (2019–2022; *n* = 1,200) and post-ChatGPT essays (2024–2025; *n* = 600). CI = confidence interval.

Primary Natural Experiment Results

Institution	Semantic Level	Post-ChatGPT	Pre-ChatGPT	<i>t</i>	df	Cohen's <i>d</i> [95% CI]
Elite Private	Word	0.16 (0.98)	-0.08 (1.00)	30.43	48291.34	0.24 [0.23, 0.26]
	Sentence	-0.06 (1.03)	0.03 (0.99)	-10.77	45835.79	-0.09 [-0.10, -0.07]
	Document	-0.10 (0.97)	0.05 (1.01)	-19.05	49240.59	-0.15 [-0.17, -0.13]
Regional Private	Word	0.12 (1.01)	-0.04 (0.99)	17.03	28077.32	0.15 [0.14, 0.17]
	Sentence	-0.14 (1.06)	0.05 (0.97)	-19.62	26524.23	-0.18 [-0.20, -0.16]
	Document	-0.06 (0.98)	0.02 (1.01)	-8.47	29110.51	-0.08 [-0.09, -0.06]
Flagship Public	Word	0.09 (1.00)	-0.06 (1.00)	28.01	131995.1	0.15 [0.14, 0.16]
	Sentence	-0.09 (1.03)	0.06 (0.97)	-28.93	127161.8	-0.15 [-0.16, -0.14]
	Document	-0.06 (0.99)	0.04 (1.01)	-17.93	134156.1	-0.09 [-0.10, -0.08]
Selective Liberal-Arts	Word	0.26 (1.02)	-0.13 (0.97)	26.54	13446.98	0.39 [0.36, 0.42]
	Sentence	-0.04 (1.05)	0.02 (0.97)	-4.22	13163.98	-0.06 [-0.09, -0.03]
	Document	-0.04 (0.96)	0.02 (1.02)	-4.58	14854.48	-0.07 [-0.10, -0.04]

Supplementary Table 7.

Natural Experiment: Semantic Diversity Changes in College Admissions Essays (Pre- and Post-ChatGPT Differences). All Welch's *t*-tests comparing standardized semantic diversity scores between pre- and post-ChatGPT essays were significant ($p < .001$). Sample periods and sizes varied across institutions: elite private – pre: 2019–2022 ($n = 47,520$), post: 2024–2025 ($n = 23,760$); regional private – pre: 2020–2022 ($n = 49,785$), post: 2024 ($n = 16,595$); flagship public – pre: 2020–2022 ($n = 92,790$), post: 2024–2025 ($n = 61,860$); selective liberal-arts – pre: 2018–2021 ($n = 14,052$), post: 2024–2025 ($n = 7,026$). The 2023 transition cycle was excluded from these analyses.

Diagnostic Analysis

To evaluate whether the post-ChatGPT essays showed stronger AI-related writing signals than the pre-ChatGPT essays, we conducted a diagnostic analysis using GPTZero⁵⁶, an AI-detection system independent of our semantic embedding measures. This analysis was not intended to identify definitively which individual essays were written with AI assistance or to estimate the true prevalence of AI use. Detector false negatives are likely, especially in admissions contexts where applicants may be motivated to avoid detection. We therefore treated GPTZero classifications as a diagnostic signal that the post-ChatGPT period contained stronger AI-related writing signals than essays from the pre-ChatGPT period.

We analyzed a balanced sample of 2,000 anonymized essays: 250 pre-ChatGPT and 250 post-ChatGPT essays from each of the four universities. Following the temporal coding used in the main analyses, 2023 transition-cycle essays were excluded. GPTZero assigns each document a classification of human, mixed, or AI. We treated AI and mixed classifications as non-human detector signals, with mixed indicating that GPTZero detected both human-like and AI-like textual signals within the essay. Pooled across institutions, the share of essays classified as AI or mixed increased from 1 of 1,000 pre-ChatGPT essays (0.1%) to 125 of 1,000 post-ChatGPT essays (12.5%; Supplementary Table 8). This increase appeared at all four institutions.

We then tested whether essays classified as AI or mixed showed the semantic pattern observed in the main analyses. Specifically, we compared all pre-ChatGPT essays with post-ChatGPT essays classified by GPTZero as AI or mixed. This detector-flagged subset reproduced the pattern of Semantic Disjunction observed in the main natural experiment data sets, and with generally much larger effect sizes: word-level diversity was higher in post-ChatGPT AI/mixed essays than in pre-ChatGPT essays, whereas sentence-level diversity and document-level distinctness were lower (Extended Data Fig. 2; Supplementary Table 8).

Across institutions, Cohen's d ranged from 0.14 to 0.62 for word-level diversity, from -1.21 to -0.34 for sentence-level diversity, and from -0.67 to -0.30 for document-level distinctness. These effects were larger than the corresponding effects in the main natural experiment analyses: across institutions, the word-level increase was on average 1.9-fold larger, and the sentence-level and document-level decreases were on average 5.3-fold and 5.8-fold larger, respectively.

These results provide diagnostic evidence that the post-ChatGPT essay samples contained stronger AI-related writing signals than the pre-ChatGPT samples. They do not establish whether any individual essay was written with AI assistance, but they support the interpretation that the natural-experiment contrast captured a meaningful shift in AI-related writing after the public release of ChatGPT.

Institution	Period	Human	Mixed	AI	AI or mixed
Elite private	Pre-ChatGPT	250	0	0	0 (0%)
Elite private	Post-ChatGPT	225	11	14	25 (10%)
Regional private	Pre-ChatGPT	249	0	1	1 (0.4%)
Regional private	Post-ChatGPT	220	12	18	30 (12%)
Flagship public	Pre-ChatGPT	250	0	0	0 (0%)
Flagship public	Post-ChatGPT	224	11	15	26 (10.4%)
Selective liberal-arts	Pre-ChatGPT	250	0	0	0 (0%)
Selective liberal-arts	Post-ChatGPT	206	21	23	44 (17.6%)

Supplementary Table 8.

Natural Experiment: GPTZero document-level classifications by institution and period. "Mixed" indicates that GPTZero detected both human-like and AI-like textual signals within the essay.

Event-Time Analysis

To assess whether our findings reflected ongoing trends rather than a discrete shift, we conducted an event-time analysis for each university. We estimated regression models using admissions cycles as indicator variables, with the 2023 cycle serving as the reference period.

Because institutional datasets covered different time spans, analytic windows varied by university: Elite Private University (2019–2025), Regional Private University (2020–2024), Flagship Public University (2020–2025). The Selective Liberal-Arts College was excluded from this analysis due to the absence of data for the 2023 reference cycle.

Extended Data Fig. 3 shows a visual summary of these results, plotting the standardized coefficients for the elite private university (top panel), the regional private university (middle panel), and the flagship public university (bottom panel). The full regression outputs are presented in Supplementary Tables 9a–c. The event-time analysis revealed largely consistent patterns across all three universities. At all universities, sentence-level diversity and document-level distinctness remained stable during the pre-ChatGPT period, with standardized coefficients close to zero. These findings supported the interpretation that the sharp post-ChatGPT changes represented discrete shifts rather than continuations of prior trends, indicating that deep conceptual homogenization is a new phenomenon emerging after the release of ChatGPT. In contrast, word-level diversity showed an upward trend at all three universities even before the release of ChatGPT, indicating that post-ChatGPT increases may partially reflect ongoing changes.

<i>Predictors</i>	Word-Level		Sentence-Level		Document-Level	
	β	95% CI	β	95% CI	β	95% CI
year2019	-0.21***	-0.24, -0.19	-0.03*	-0.06, -0.01	0.01	-0.02, 0.03
year2020	-0.21***	-0.24, -0.19	0.00	-0.02, 0.03	0.00	-0.02, 0.03
year2021	-0.10***	-0.12, -0.07	-0.02	-0.04, 0.01	0.01	-0.02, 0.03
year2022	-0.04**	-0.06, -0.01	0.01	-0.02, 0.03	0.03*	0.00, 0.05
year2024	0.05***	0.02, 0.08	-0.09***	-0.12, -0.06	-0.11***	-0.14, -0.09
year2025	0.15***	0.12, 0.17	-0.10***	-0.13, -0.08	-0.16***	-0.19, -0.14
<i>N</i>	83,160		83,160		83,160	
<i>R</i> ²	.015		.002		.005	

Supplementary Table 9a.

Natural Experiment: Event-Time Analysis for Semantic Diversity by Embedding Levels (Elite Private University). The reference group was the 2023 cycle. CI = confidence interval. * $p < .05$.

** $p < .01$. *** $p < .001$.

<i>Predictors</i>	Word-Level		Sentence-Level		Document-Level	
	β	95% CI	β	95% CI	β	95% CI
year2020	-0.21***	-0.23, -0.19	-0.02	-0.04, 0.01	-0.01	-0.03, 0.02
year2021	-0.05***	-0.07, -0.02	0.06***	0.04, 0.08	-0.03**	-0.06, -0.01
year2022	0.02	-0.01, 0.04	0.07***	0.05, 0.10	0.02	-0.00, 0.04
year2024	0.07***	0.05, 0.10	-0.14***	-0.17, -0.12	-0.08***	-0.10, -0.06
<i>N</i>	82,975		82,967		82,975	
<i>R</i> ²	.009		.006		.001	

Supplementary Table 9b.

Natural Experiment: Event-Time Analysis for Semantic Diversity by Embedding Levels

(Regional Private University). The reference group was the 2023 cycle. CI = confidence interval.

** $p < .01$. *** $p < .001$.

<i>Predictors</i>	Word-Level		Sentence-Level		Document-Level	
	β	95% CI	β	95% CI	β	95% CI
year2020	-0.08***	-0.10, -0.07	-0.02*	-0.03, -0.00	-0.07***	-0.09, -0.06
year2021	-0.06***	-0.08, -0.05	-0.05***	-0.06, -0.03	-0.07***	-0.08, -0.05
year2022	-0.04***	-0.06, -0.03	-0.00	-0.02, 0.01	0.01	-0.01, 0.02
year2024	0.06***	0.04, 0.07	-0.13***	-0.14, -0.11	-0.08***	-0.09, -0.06
year2025	0.11***	0.09, 0.12	-0.22***	-0.24, -0.21	-0.20***	-0.21, -0.18
<i>N</i>	185,580		185,579		185,580	
<i>R</i> ²	.005		.006		.005	

Supplementary Table 9c.

Natural Experiment: Event-Time Analysis for Semantic Diversity by Embedding Levels

(Flagship Public University). The reference group was the 2023 cycle. CI = confidence interval.

* $p < .05$. *** $p < .001$.

COVID-19 Pandemic Confound Analysis

To rule out the COVID-19 pandemic as an alternative explanation, we conducted separate regression analyses for each university. These models distinguished potential pandemic-related effects from post-ChatGPT changes by using the main pandemic period (2021–2022) as a reference. The results are presented in Supplementary Tables 10a–d for each university. Our regression analyses confirmed that the semantic disjunction was specific to the post-ChatGPT period and not attributable to lingering effects of the pandemic.

At all four universities, the pandemic period was associated with a modest increase in word-level diversity compared to the pre-COVID baseline. Critically, this period showed no decline in sentence or document-level diversity. In fact, sentence-level diversity at the regional private university and document-level distinctness at the flagship public university were slightly higher during the pandemic than before it. This trend contrasts with the sharp declines observed post-ChatGPT, confirming that the deep conceptual homogenization was a new phenomenon and not a pandemic-related confound.

<i>Predictors</i>	Word-Level		Sentence-Level		Document-Level	
	β	95% CI	β	95% CI	β	95% CI
Pre-COVID/Pre-ChatGPT	-0.14***	-0.16, -0.13	-0.01	-0.03, 0.01	-0.01	-0.03, 0.01
Post-COVID/Post-ChatGPT	0.17***	0.15, 0.18	-0.09***	-0.11, -0.07	-0.15***	-0.17, -0.14
<i>N</i>	71,280		71,280		71,280	
<i>R</i> ²	.016		.002		.005	

Supplementary Table 10a.

Natural Experiment: Regression Results Predicting Semantic Diversity from COVID- and ChatGPT-Related Periods (Elite Private University). The regression models compared the pre-COVID/pre-ChatGPT era (2019–20 cycles; $n = 23,760$) and the post-COVID/post-ChatGPT era (2024–25 cycles; $n = 23,760$) with the main COVID-affected period before ChatGPT's public release (2021–22 cycles; $n = 23,760$). The 2021 and 2022 periods thus served as the reference to isolate the pandemic effect. The 2023 transition cycle was excluded from analysis. CI = confidence interval. *** $p < .001$.

<i>Predictors</i>	Word-Level		Sentence-Level		Document-Level	
	β	95% CI	β	95% CI	β	95% CI
Pre-COVID/Pre-ChatGPT	-0.19***	-0.21, -0.17	-0.08***	-0.10, -0.06	0.00	-0.02, 0.02
Post-COVID/Post-ChatGPT	0.09***	0.07, 0.11	-0.21***	-0.23, -0.19	-0.07***	-0.09, -0.06
<i>N</i>	66,380		66,374		66,380	
<i>R</i> ²	.011		.007		.001	

Supplementary Table 10b.

Natural Experiment: Regression Results Predicting Semantic Diversity from COVID- and ChatGPT-Related Periods (Regional Private University). The regression models compared the pre-COVID/pre-ChatGPT era (2020 cycle; $n = 16,595$) and the post-COVID/post-ChatGPT era (2024 cycle; $n = 16,595$) with the main COVID-affected period before ChatGPT's public release (2021–22 cycles; $n = 33,190$). The 2021 and 2022 periods served as the reference to isolate the pandemic effect. The 2023 transition cycle was excluded from analysis. CI = confidence interval.

*** $p < .001$.

<i>Predictors</i>	Word-Level		Sentence-Level		Document-Level	
	β	95% CI	β	95% CI	β	95% CI
Pre-COVID/Pre-ChatGPT	-0.03***	-0.04, -0.02	0.01	-0.01, 0.02	-0.04***	-0.06, -0.03
Post-COVID/Post-ChatGPT	0.14***	0.12, 0.15	-0.15***	-0.16, -0.14	-0.11***	-0.12, -0.10
<i>N</i>	154,650		154,649		154,650	
<i>R</i> ²	.005		.006		.002	

Supplementary Table 10c.

Natural Experiment: Regression Results Predicting Semantic Diversity from COVID- and ChatGPT-Related Periods (Flagship Public University). The regression models compared the pre-COVID/pre-ChatGPT era (2020 cycle; $n = 30,930$) and the post-COVID/post-ChatGPT era (2024–25 cycle; $n = 61,860$) with the main COVID-affected period before ChatGPT's public release (2021–22 cycles; $n = 61,860$). The 2021 and 2022 periods served as the reference to isolate the pandemic effect. The 2023 transition cycle was excluded from analysis. CI = confidence interval. *** $p < .001$.

<i>Predictors</i>	Word-Level		Sentence-Level		Document-Level	
	β	95% CI	β	95% CI	β	95% CI
Pre-COVID/Pre-ChatGPT	-0.13***	-0.17, -0.09	0.02	-0.01, 0.06	-0.02	-0.06, 0.01
Post-COVID/Post-ChatGPT	0.29***	0.25, 0.33	-0.04*	-0.09, -0.00	-0.08***	-0.12, -0.04
<i>N</i>	21,078		21,078		21,078	
<i>R</i> ²	.035		.001		.001	

Supplementary Table 10d.

Natural Experiment: Regression Results Predicting Semantic Diversity from COVID- and ChatGPT-Related Periods (Selective Liberal-Arts College). The regression models compared the pre-COVID/pre-ChatGPT era (2018–20 cycle; $n = 10,539$) and the post-COVID/post-ChatGPT era (2024–25 cycle; $n = 7,026$) with the main COVID-affected period before ChatGPT's public release (2021 cycle; $n = 3,513$). The 2021 period served as the reference to isolate the pandemic effect. The 2023 transition cycle was excluded from analysis. CI = confidence interval. *** $p < .001$.

Collective Word-Level Diversity Across Essays

To address whether the observed disjunction reflected a mismatch between within-essay and between-essay units of analysis, we conducted an additional robustness check measuring collective word-level diversity across essays. In the primary analysis, word-level diversity was computed within each essay, whereas document-level distinctness was computed across essays within the same cohort. This raised the possibility that the apparent disjunction could partly reflect differences in unit of analysis rather than a substantive divergence between lexical and conceptual semantics. This robustness check therefore tested whether post-ChatGPT cohorts also showed greater word-level diversity when essays served as the sampling unit.

For each institution-year-period cell, we repeatedly sampled essays without replacement. Within each resample, each selected essay contributed one randomly selected sentence. We then pooled the word-level BERT embeddings from those selected sentences and computed collective word-level diversity using the same DSI procedure as in the primary word-level analysis. Higher values indicate greater semantic diversity among words contributed by different essays in the sampled cohort. We repeated this procedure 5,000 times for each cell across nested essay sample sizes of 2, 4, 8, 16, 32, 64, 128, and 256 essays per resample.

For statistical summaries, we first averaged the 5,000 resamples within each institution-year-period-sample-size cell, so that repeated resampling draws were not treated as independent observations. We then standardized the resulting collective word-level diversity values within each institution and estimated institution-specific models predicting standardized collective word-level diversity from ChatGPT period, log2-transformed essay sample size, and their interaction. From these models, we computed the estimated post-minus-pre contrast at each essay sample size.

The results showed that the post-ChatGPT increase in word-level diversity was not limited to the original within-essay metric. Post-ChatGPT cohorts showed higher collective word-level diversity across essays than pre-ChatGPT cohorts in every institution, and this pattern was stable across essay sample sizes from 2 to 256 essays per resample (Extended Data Fig. 4).

5 All 32 institution-by-sample-size contrasts were positive, with standardized post-minus-pre differences ranging from 0.088 to 0.297; all p values were below .027. The estimated ranges were positive in each institution: elite private university, 0.147 to 0.186; regional private university, 0.186 to 0.189; flagship public university, 0.088 to 0.117; and selective liberal-arts college, 0.243 to 0.297.

10 This pattern is conceptually plausible because lexical variation and document-level thematic distinctness need not move together. AI-assisted revision may help applicants rephrase sentences, substitute words, or elaborate descriptions, increasing word-level variation even as essays converge on similar higher-order narratives and themes. For example, two essays may both present a familiar narrative of overcoming adversity while using different words to describe
15 the challenge, effort, and growth. In such cases, word-level diversity can increase while document-level distinctness decreases, because the lexical surface becomes more varied while the underlying conceptual trajectory becomes more similar.

20 This robustness check indicates that the word-level increase was not simply a within-essay lexical elaboration effect. Even when essays were treated as the sampling unit and each essay contributed only one randomly selected sentence, post-ChatGPT cohorts remained more diverse at the word level. Thus, the disjunction between increased word-level diversity and decreased document-level distinctness was not explained by comparing a within-essay word-level metric with a between-essay document-level metric. Rather, word-level diversity increased both within essays and across essays, while document-level distinctness decreased across essays.

Models with Control Variables

To ensure our findings were not an artifact of changing applicant demographics or essay length, we re-ran our main analyses, including several control variables. These analyses were conducted separately for all four universities. The availability of control variables varied across institutions.

5 As shown in Supplementary Tables 11a–d, the key post-ChatGPT effects on semantic diversity remained significant across all universities, confirming the robustness of our main findings on semantic disjunction.

<i>Predictors</i>	Word-Level		Sentence-Level		Document-Level	
	β	95% CI	β	95% CI	β	95% CI
Post-ChatGPT	0.23***	0.21, 0.24	-0.12***	-0.13, -0.10	-0.16***	-0.18, -0.14
Word Count	0.11***	0.10, 0.12	-0.07***	-0.08, -0.07	-0.08***	-0.08, -0.07
Sentence Count	0.03***	0.03, 0.04	0.49***	0.49, 0.50	0.13***	0.12, 0.13
Controls	Yes		Yes		Yes	
<i>N</i>	71,280		71,280		71,280	
<i>R</i> ²	.068		.228		.026	

Supplementary Table 11a.

Natural Experiment: Post-ChatGPT Differences in Semantic Diversity Adjusted for Control Variables (Elite Private University). Control variables include sex, race, ethnicity, international status, and English spoken at home. CI = confidence interval. *** $p < .001$.

<i>Predictors</i>	Word-Level		Sentence-Level		Document-Level	
	β	95% CI	β	95% CI	β	95% CI
Post-ChatGPT	0.18***	0.16, 0.20	-0.18***	-0.20, -0.17	-0.10***	-0.12, -0.08
Word Count	0.30***	0.29, 0.31	-0.02***	-0.03, -0.01	-0.20***	-0.21, -0.19
Sentence Count	0.02**	0.01, 0.02	0.54***	0.53, 0.54	0.08***	0.07, 0.09
Controls	Yes		Yes		Yes	
<i>N</i>	66,380		66,374		66,380	
<i>R</i> ²	.136		.285		.046	

Supplementary Table 11b.

Natural Experiment: Post-ChatGPT Differences in Semantic Diversity Adjusted for Control Variables (Regional Private University). Control variables include sex, race, ethnicity, and international status. CI = confidence interval. *** $p < .001$.

<i>Predictors</i>	Word-Level		Sentence-Level		Document-Level	
	β	95% CI	β	95% CI	β	95% CI
Post-ChatGPT	0.16***	0.15, 0.17	-0.17***	-0.17, -0.16	-0.09***	-0.10, -0.08
Word Count	0.19***	0.18, 0.19	-0.06***	-0.06, -0.05	-0.07***	-0.08, -0.07
Sentence Count	-0.05***	-0.06, -0.05	0.46***	0.45, 0.46	0.04***	0.04, 0.05
Controls	Yes		Yes		Yes	
<i>N</i>	154,650		154,649		154,650	
<i>R</i> ²	.078		.200		.010	

Supplementary Table 11c.

Natural Experiment: Post-ChatGPT Differences in Semantic Diversity Adjusted for Control Variables (Flagship Public University). Control variables include sex, race, and ethnicity. CI = confidence interval. *** $p < .001$.

<i>Predictors</i>	Word-Level		Sentence-Level		Document-Level	
	β	95% CI	β	95% CI	β	95% CI
Post-ChatGPT	0.29***	0.27, 0.32	-0.12***	-0.15, -0.09	-0.09***	-0.12, -0.06
Word Count	0.23***	0.22, 0.25	-0.02***	-0.04, -0.01	-0.08***	-0.10, -0.07
Sentence Count	-0.04***	-0.05, -0.03	0.47***	0.45, 0.48	0.04***	0.03, 0.06
Controls	Yes		Yes		Yes	
<i>N</i>	21,078		21,078		21,078	
<i>R</i> ²	.110		.223		.011	

Supplementary Table 11d.

Natural Experiment: Post-ChatGPT Differences in Semantic Diversity Adjusted for Control Variables (Selective Liberal-Arts College). Control variables include race and ethnicity. CI = confidence interval.

5 *** $p < .001$.

Racial and Linguistic Subgroup Analyses

To assess the generalizability and variability of semantic disjunction by demographic group, we ran separate multiple regression models for each semantic level. Each model included main effects for ChatGPT release and a demographic variable (e.g., racial/ethnic subgroup or linguistic background), as well as their interaction. Generalizability was evaluated by the simple effect of ChatGPT release within each subgroup, with Holm correction for multiple comparisons.

Variability was assessed by the interaction term: a significant interaction indicated that semantic changes differed from the White American reference group. A stronger semantic disjunction effect was defined by greater word-level diversity accompanied by steeper declines in sentence-level diversity and document-level distinctness.

<i>Predictors</i>	Word-Level		Sentence-Level		Document-Level	
	β	95% CI	β	95% CI	β	95% CI
Post-ChatGPT	0.16***	0.13, 0.18	-0.07***	-0.09, -0.04	-0.12***	-0.15, -0.09
[Asian American]	0.31***	0.29, 0.34	0.07***	0.04, 0.09	-0.12***	-0.14, -0.09
[Black/African American]	-0.41***	-0.44, -0.37	-0.07***	-0.11, -0.03	0.05*	0.01, 0.08
[Hispanic/Latine]	-0.20***	-0.23, -0.17	-0.01	-0.04, 0.02	-0.11***	-0.14, -0.08
[Indigenous Peoples]	-0.21	-0.42, 0.01	-0.13	-0.35, 0.09	0.05	-0.17, 0.27
[International]	0.11***	0.08, 0.14	0.18***	0.15, 0.21	0.18***	0.15, 0.21
[Missing]	0.41***	0.27, 0.56	0.14	-0.01, 0.29	-0.01	-0.15, 0.14
[Prefer not to disclose]	0.16***	0.10, 0.21	0.05	-0.01, 0.11	-0.03	-0.09, 0.02
[Two or more races]	0.06**	0.02, 0.09	0.05*	0.01, 0.09	-0.02	-0.06, 0.01
Post-ChatGPT $\times$ [Asian American]	0.07***	0.03, 0.11	-0.04	-0.08, 0.01	-0.02	-0.06, 0.02
Post-ChatGPT $\times$ [Black/African American]	0.09*	0.02, 0.16	-0.11**	-0.18, -0.04	-0.08*	-0.15, -0.01
Post-ChatGPT $\times$ [Hispanic/Latine]	0.05	-0.00, 0.10	-0.06*	-0.11, -0.01	0.00	-0.05, 0.05
Post-ChatGPT $\times$ [Indigenous Peoples]	-0.08	-0.43, 0.27	-0.03	-0.39, 0.32	0.19	-0.17, 0.54
Post-ChatGPT $\times$ [International]	0.15***	0.10, 0.20	-0.07**	-0.13, -0.02	-0.11***	-0.16, -0.06
Post-ChatGPT $\times$ [Missing]	-0.20	-0.85, 0.45	-0.34	-1.01, 0.33	-0.60	-1.27, 0.07
Post-ChatGPT $\times$ [Prefer not to disclose]	0.12***	0.05, 0.20	0.04	-0.03, 0.12	-0.02	-0.09, 0.06
Post-ChatGPT $\times$ [Two or more races]	0.06	-0.01, 0.12	-0.01	-0.08, 0.05	-0.03	-0.09, 0.04
<i>N</i>	71,280		71,280		71,280	
<i>R</i> ²	.050		.006		.012	

Supplementary Table 12a.

Natural Experiment: Semantic Diversity Scores Before and After ChatGPT Release Across Racial and Ethnic Subgroups (Elite Private University). The reference group was the White American group. CI = confidence interval. * $p < .05$. ** $p < .01$. *** $p < .001$.

<i>Predictors</i>	Word-Level		Sentence-Level		Document-Level	
	β	95% CI	β	95% CI	β	95% CI
Post-ChatGPT	0.07***	0.04, 0.10	-0.12***	-0.15, -0.08	-0.07***	-0.10, -0.03
[Asian American]	0.15***	0.12, 0.17	-0.04**	-0.06, -0.01	-0.13***	-0.15, -0.10
[Black/African American]	-0.20***	-0.23, -0.16	-0.03*	-0.07, -0.00	0.14***	0.11, 0.17
[Hispanic/Latine]	-0.15***	-0.20, -0.09	-0.05	-0.10, 0.01	-0.06*	-0.12, -0.00
[Indigenous Peoples]	-0.34***	-0.41, -0.27	-0.12**	-0.19, -0.05	0.03	-0.04, 0.10
[International]	0.15***	0.12, 0.18	0.25***	0.22, 0.28	0.31***	0.28, 0.34
[Two or more races]	0.03	-0.00, 0.06	0.01	-0.02, 0.04	-0.04*	-0.07, -0.01
[Missing]	-0.51***	-0.54, -0.48	-0.21***	-0.24, -0.18	-0.17***	-0.20, -0.14
Post-ChatGPT $\times$ [Asian American]	0.17***	0.12, 0.22	-0.01	-0.06, 0.05	-0.00	-0.05, 0.05
Post-ChatGPT $\times$ [Black/African American]	0.10***	0.04, 0.16	-0.12***	-0.18, -0.06	-0.01	-0.07, 0.05
Post-ChatGPT $\times$ [Hispanic/Latine]	-0.07	-0.19, 0.05	-0.09	-0.21, 0.03	0.02	-0.10, 0.14
Post-ChatGPT $\times$ [Indigenous Peoples]	0.05	-0.09, 0.19	-0.10	-0.24, 0.04	-0.04	-0.19, 0.10
Post-ChatGPT $\times$ [International]	0.12***	0.07, 0.17	-0.30***	-0.35, -0.25	-0.13***	-0.19, -0.08
Post-ChatGPT $\times$ [Two or more races]	0.07*	0.00, 0.13	-0.02	-0.08, 0.04	0.02	-0.04, 0.08
Post-ChatGPT $\times$ [Missing]	0.12***	0.06, 0.19	-0.02	-0.09, 0.05	0.05	-0.02, 0.12
<i>N</i>	66,380		66,374		66,380	
<i>R</i> ²	.041		.018		.020	

Supplementary Table 12b.

Natural Experiment: Semantic Diversity Scores Before and After ChatGPT Release Across Racial and Ethnic Subgroups (Regional Private University). The reference group was the White American group. CI = confidence interval. * $p < .05$. ** $p < .01$. *** $p < .001$.

<i>Predictors</i>	Word-Level		Sentence-Level		Document-Level	
	β	95% CI	β	95% CI	β	95% CI
Post-ChatGPT	0.12***	0.10, 0.13	-0.08***	-0.09, -0.06	-0.08***	-0.10, -0.07
[Asian American]	0.27***	0.25, 0.29	0.13***	0.11, 0.15	0.04**	0.01, 0.06
[Black/African American]	-0.35***	-0.37, -0.32	0.03**	0.01, 0.06	0.18***	0.15, 0.20
[Hispanic/Latine]	-0.30***	-0.32, -0.28	0.03**	0.01, 0.05	0.06***	0.04, 0.08
[Indigenous Peoples]	-0.24**	-0.41, -0.07	-0.12	-0.30, 0.05	0.17	-0.01, 0.34
[International]	0.27***	0.25, 0.29	0.23***	0.21, 0.26	0.10***	0.07, 0.12
[Two or more races]	0.03	-0.00, 0.07	0.07***	0.04, 0.11	0.10***	0.07, 0.14
[Missing]	-0.00	-0.04, 0.04	0.08***	0.04, 0.12	0.10***	0.07, 0.14
Post-ChatGPT $\times$ [Asian American]	0.19***	0.16, 0.23	-0.07***	-0.10, -0.04	-0.02	-0.05, 0.02
Post-ChatGPT $\times$ [Black/African American]	0.09***	0.05, 0.12	-0.19***	-0.23, -0.16	-0.09***	-0.13, -0.05
Post-ChatGPT $\times$ [Hispanic/Latine]	0.05***	0.02, 0.08	-0.12***	-0.15, -0.09	-0.04*	-0.07, -0.01
Post-ChatGPT $\times$ [Indigenous Peoples]	0.26*	0.02, 0.50	0.07	-0.17, 0.32	-0.02	-0.27, 0.23
Post-ChatGPT $\times$ [International]	0.09***	0.05, 0.13	-0.26***	-0.30, -0.22	0.12***	0.08, 0.15
Post-ChatGPT $\times$ [Two or more races]	0.09***	0.04, 0.14	-0.08**	-0.13, -0.02	-0.06*	-0.12, -0.01
Post-ChatGPT $\times$ [Missing]	0.10**	0.03, 0.16	-0.07*	-0.13, -0.00	-0.00	-0.07, 0.07
<i>N</i>	154,650		154,649		154,650	
<i>R</i> ²	.048		.010		.005	

Supplementary Table 12c.

Natural Experiment: Semantic Diversity Scores Before and After ChatGPT Release Across Racial and Ethnic Subgroups (Flagship Public University). The reference group was the White American group. CI = confidence interval. * $p < .05$. ** $p < .01$. *** $p < .001$.

<i>Predictors</i>	Word-Level		Sentence-Level		Document-Level	
	β	95% CI	β	95% CI	β	95% CI
Post-ChatGPT	0.20***	0.16, 0.25	-0.04	-0.09, 0.01	0.05*	0.00, 0.10
[Asian American]	0.09**	0.03, 0.15	0.13***	0.07, 0.19	-0.03	-0.09, 0.03
[Black/African American]	-0.52***	-0.59, -0.44	-0.12**	-0.19, -0.05	0.09*	0.01, 0.16
[Hispanic/Latine]	-0.21***	-0.27, -0.15	0.04	-0.02, 0.10	-0.06	-0.12, 0.01
[Indigenous Peoples]	-0.21	-0.55, 0.14	0.27	-0.09, 0.62	0.18	-0.18, 0.54
[International]	0.09***	0.05, 0.13	0.31***	0.27, 0.35	0.29***	0.25, 0.33
[Two or more races]	-0.07	-0.16, 0.01	0.04	-0.04, 0.13	-0.06	-0.15, 0.03
[Missing]	0.12	-0.01, 0.25	0.08	-0.05, 0.20	-0.02	-0.15, 0.11
Post-ChatGPT $\times$ [Asian American]	0.12*	0.00, 0.23	-0.15*	-0.27, -0.03	-0.01	-0.13, 0.11
Post-ChatGPT $\times$ [Black/African American]	0.12	-0.03, 0.26	-0.08	-0.23, 0.07	-0.12	-0.27, 0.03
Post-ChatGPT $\times$ [Hispanic/Latine]	-0.07	-0.19, 0.04	-0.09	-0.21, 0.03	0.05	-0.07, 0.17
Post-ChatGPT $\times$ [Indigenous Peoples]	-0.11	-1.02, 0.81	-0.06	-1.00, 0.88	0.17	-0.77, 1.11
Post-ChatGPT $\times$ [International]	0.29***	0.22, 0.35	-0.16***	-0.23, -0.10	-0.39***	-0.46, -0.32
Post-ChatGPT $\times$ [Two or more races]	-0.05	-0.21, 0.12	-0.09	-0.25, 0.08	0.05	-0.11, 0.22
Post-ChatGPT $\times$ [Missing]	0.11	-0.11, 0.32	0.12	-0.10, 0.34	-0.11	-0.33, 0.11
<i>N</i>	21,078		21,078		21,078	
<i>R</i> ²	.067		.015		.012	

Supplementary Table 12d.

Natural Experiment: Semantic Diversity Scores Before and After ChatGPT Release Across Racial and Ethnic Subgroups (Selective Liberal-Arts College). The reference group was the White American group. CI = confidence interval. * $p < .05$. ** $p < .01$. *** $p < .001$.

<i>Predictors</i>	Word-Level		Sentence-Level		Document-Level	
	β	95% CI	β	95% CI	β	95% CI
Post-ChatGPT	0.24***	0.22, 0.26	-0.06***	-0.08, -0.04	-0.14***	-0.16, -0.12
[Non-English at home]	0.01	-0.01, 0.03	0.10***	0.08, 0.12	0.04***	0.02, 0.06
Post-ChatGPT $\times$ [Non-English at home]	0.07***	0.03, 0.11	-0.06***	-0.10, -0.03	-0.05*	-0.08, -0.01
<i>N</i>	64,550		64,550		64,550	
<i>R</i> ²	.015		.003		.005	

Supplementary Table 12e.

Natural Experiment: Semantic Diversity Scores Before and After ChatGPT Release by Language

Background (Elite Private University). The reference group was applicants from English-

speaking households. CI = confidence interval. * $p < .05$. *** $p < .001$.

Investigating Mechanisms: LLM Decoupling of Lexical and Conceptual Semantics

Controlled Experiment 1: Essay Generation and Matching Procedures. We first aggregated admissions essays from regional private and selective liberal-arts universities for which corresponding essay prompts were available. We filtered the dataset to include only essays submitted prior to the 2023 admissions cycle to ensure the human baselines were free of LLM influence. From this pre-LLM corpus, we stratified by university and essay topic, randomly sampling 100 human essays per prompt separately for each institution. This selection yielded a balanced human baseline covering 7 types of essay prompts from the regional private university (2020–2022) and 8 types from the selective liberal-arts college (2018–2020).

For each selected human essay, we generated a synthetic counterpart using GPT-5-mini (gpt-5-mini-2025-08-07). To ensure a precise comparison, the model was provided with the identical prompt answered by the human applicant, matching the specific word count of the human reference essay. The system instruction was defined as: *"You are an excellent college admissions essay writer. Write a compelling, personal essay of about {target_word_count} words. Be creative, authentic, reflective, and specific."* The Welch two-sample *t*-test showed a significant but small difference in word counts, with human essays ($M = 598.20$, $SD = 91.92$) averaging approximately 34 fewer words than their AI counterparts ($M = 632.47$, $SD = 142.43$; $t(2563.06) = -7.83$, $p < .001$, Cohen's $d = -0.29$).

To test the "disjunction" hypothesis, that AI weakens the link between surface-level semantic diversity and deeper conceptual novelty in human-written text, we analyzed the association between semantic levels (e.g., word vs. document) across conditions. For the human versus AI-generated comparison, we compared Pearson correlation coefficients between semantic levels.

Real-World Replications in Natural Experiments: Pre- vs. Post-ChatGPT. Results

from our large-scale natural experiment datasets further supported the patterns observed in the matched comparisons (Supplementary Table 13a). We fitted regression models predicting word-level diversity from sentence- or document-level metrics, including an interaction term for the post-ChatGPT period to test if the strength of these associations was significantly different across periods.

Across all four institutions, document-level distinctness was positively associated with word-level diversity ($\beta = 0.15$ to 0.18 , $ps < .001$; see Supplementary Table 13b) in pre-ChatGPT essays. However, the association became weaker in the post-ChatGPT period, as indicated by negative-directional interaction effects that were significant at three of the universities ($\beta = -0.03$ to -0.05 , $ps < .001$), though non-significant at the elite private university ($\beta = -0.01$, $p = .164$). The changes in the relationship between word- and sentence-level diversity following the ChatGPT release were mixed (Supplementary Table 13c): while the association weakened for the regional private ($\beta = -0.05$, $p < .001$) and flagship public universities ($\beta = -0.05$, $p < .001$), it strengthened for the elite private university ($\beta = 0.03$, $p < .001$), and was non-significant for the selective liberal-arts college ($\beta = 0.01$, $p = .427$). These results provide ecological evidence from real-world writing that the link between surface-level semantic diversity and conceptual novelty has weakened following the ChatGPT release.

	Word-Level	Sentence-Level	Document-Level
Word-Level	–	0.430***	0.117***
Sentence-Level	0.051*	–	0.076**
Document-Level	-0.283***	0.054*	–

Supplementary Table 13a.

Controlled Experiment 1: Correlations Among Semantic Metrics in Human and GPT-Generated Essays. Upper triangle reports correlations for human-written essays ($n = 1,500$), and lower triangle reports correlations for GPT-generated essays ($n = 1,500$). * $p < .05$. ** $p < .01$. *** $p < .001$.

<i>Predictors</i>	Elite Private		Regional Private		Flagship Public		Selective Liberal-Arts	
	β	95% CI	β	95% CI	β	95% CI	β	95% CI
Document-Level Distinctness	0.15***	0.14, 0.16	0.15***	0.14, 0.15	0.18***	0.17, 0.18	0.17***	0.15, 0.18
Post-ChatGPT	0.26***	0.24, 0.28	0.16***	0.15, 0.18	0.16***	0.15, 0.17	0.40***	0.37, 0.42
Document-Level Distinctness $\times$ Post-ChatGPT	-0.01	-0.03, 0.00	0.03***	-0.05, -0.01	0.04***	-0.05, -0.03	0.05***	-0.08, -0.02
<i>N</i>	71,280		66,380		154,650		21,078	
<i>R</i> ²	.034		.023		.032		.057	

Supplementary Table 13b.

Natural Experiments: Changes in the Association Between Word-Level Diversity and Document-Level Distinctness Following the Release of ChatGPT. In this regression model, β represents the standardized coefficients of independent variables predicting word-level diversity.

CI = confidence interval. * $p < .05$. *** $p < .001$.

<i>Predictors</i>	Elite Private		Regional Private		Flagship Public		Selective Liberal-Arts	
	β	95% CI	β	95% CI	β	95% CI	β	95% CI
Sentence-Level Diversity	0.38***	0.37, 0.39	0.45***	0.44, 0.46	0.40***	0.40, 0.41	0.40***	0.39, 0.42
Post-ChatGPT	0.27***	0.26, 0.29	0.23***	0.21, 0.25	0.20***	0.19, 0.21	0.41***	0.39, 0.44
Sentence-Level Diversity $\times$ Post-ChatGPT	0.03***	0.02, 0.05	-0.05***	-0.06, -0.03	-0.05***	-0.05, -0.04	0.01	-0.01, 0.04
<i>N</i>	71,280		66,374		154,649		21,078	
<i>R</i> ²	.163		.195		.152		.198	

Supplementary Table 13c.

Natural Experiments: Changes in the Association Between Word-Level Diversity and Sentence-Level Diversity Following the Release of ChatGPT. In this regression model, β represents the standardized coefficients of independent variables predicting word-level diversity. CI = confidence interval. *** $p < .001$.

Resolving Paradoxical Effects: Post-ChatGPT Essays Were Rated as More Creative Despite Conceptual Homogenization

Assessment Analysis. To understand why post-ChatGPT essays receive higher creativity ratings despite conceptual homogenization, we utilized creativity ratings from human experts and LLMs to identify potential evaluative biases. We estimated multiple linear regression models predicting creativity ratings using the three semantic metrics simultaneously. We then applied linear hypothesis tests to compare standardized regression coefficients, determining if word-level diversity remained the disproportionate predictor of creativity scores for both humans and the expert-tuned model.

Human Experts. We recruited a panel of 22 experts specializing in creativity research to evaluate the creativity of 440 college admissions essays drawn from three different universities: two private institutions in our dataset (one elite, one regional) and one flagship public university not included in our dataset. All of these essays were submitted before the public release of ChatGPT in November 2022.

To ensure coverage across the full range of creativity, essays were first screened and stratified using DSI³². Essays with extremely low DSI scores (more than 5 standard deviations below the mean), typically reflecting extraction errors or incomplete submissions, were excluded. The remaining essays were stratified into five DSI-based categories, from which approximately 30 essays per category were sampled within each university, yielding the final set of 440 essays evaluated by experts.

The expert panel comprised university faculty actively conducting creativity research ($n = 11$), postdoctoral scholars ($n = 4$), and Ph.D. students ($n = 7$). As part of the rating procedure, all experts first evaluated a common set of 10 essays and then met to discuss rating criteria and align interpretations before independently scoring the remaining essays. Twelve experts each

rated two essay sets of $n = 40$ essays, and the remaining ten experts rated one set each. Experts were asked to read all allocated essays in each set ($n = 40$) before rating creativity. Each expert independently rated creativity on a 5-point Likert scale (1 = low creativity, 5 = high creativity), with each essay evaluated by at least three experts.

Although admissions essays often invite distinctive self-expression, they function primarily as persuasive texts, posing a question about whether creativity-relevant variation can be reliably assessed within this genre. To address this question, we tested whether expert ratings varied meaningfully across essays and converged reliably across raters. The ratings supported both criteria. Expert ratings spanned the full 1–5 scale ($M = 2.96$, $SD = 1.02$, $IQR = 2.00\text{--}3.75$), indicating substantial variation across essays. This variation was also reliable across raters: inter-rater agreement was excellent, $ICC(1,k) = 0.92$. This combination suggests that the ratings captured a strong, reliable and discriminating signal of creativity, rather than noise or range-restricted impressions. As convergent evidence, ratings from a base GPT-5-mini model correlated strongly with expert creativity ratings in the held-out test set ($r = .67$, $n = 111$), suggesting that the expert ratings also aligned with an independent evaluative system. Thus, while personal statements are not ideal creativity-assessment instruments, creativity experts could reliably distinguish essays perceived as more versus less creative in this real-world writing context.

To further refine the dataset, we excluded 70 essays with higher inter-rater disagreement ($SD > 1.0$), yielding 370 essays with mean creativity scores transformed into percentiles (0 = lowest creativity, 1 = highest creativity).

Expert-Tuned GPT-4.1-mini Model. We fine-tuned OpenAI’s GPT-4.1-mini model (gpt-4.1-mini-2025-04-14) using the expert creativity ratings to generate automated creativity ratings of essays (see *Human Experts* above). We selected the GPT-4.1-mini model as the base

for fine-tuning, to balance performance and cost; it was also the newest GPT variant available for fine-tuning at the time of analysis. We converted the average creativity ratings from human experts into percentile scores and expert-tuned the model to predict them. The dataset was partitioned into a training set (70%, $n = 259$) and a held-out test set (30%, $n = 111$). We further allocated 10% ($n = 26$) of the training set as a validation subset. We used the default fine-tuning parameters, which were automatically determined by the dataset size (Epochs = 3; Batch size = 1; LR multiplier = 2; Seed = 1).

Base GPT-5-mini Model. To serve as a baseline for automated assessment, we utilized the base GPT-5-mini model (gpt-5-mini-2025-08-07), accessed via OpenAI’s API. This model was selected because it was the newest GPT variant available at the time of analysis that offered state-of-the-art performance at a reasonable computational cost.

Controlled Experiment 2: No Systematic Influence of Joint Evaluation. To determine if the observed lexical bias from the assessment analysis was an artifact of evaluating essays one at a time (i.e., single evaluation), we tested whether providing a comparative context (i.e., joint evaluation) would mitigate the reliance on surface-level semantic diversity in creativity evaluation. We conducted a controlled experiment using both the expert-tuned GPT-4.1-mini and Base GPT-5-mini models. We modified the LLM prompt to include a “*reference cohort*” of previously written essays, increasing the number of examples exponentially from 0 to 128 (2^n). The prompt instructed the model to first read these reference essays to calibrate its internal baseline for creativity before assigning a score to the target essay. This setup was intended to mirror the evaluative setting of our human experts who read approximately 50 essays before providing ratings.

We utilized the aforementioned $N = 370$ human-written essays rated by 22 experts. The test dataset ($n = 111$) was used for final evaluation, while the reference essays were randomly

selected exclusively from the training dataset ($n = 259$). This selection strategy ensured that the reference essays and test items remained independent of each other, preventing any compounding effects on evaluation. These reference examples were provided without human scores to ensure the model was evaluating the comparative distribution of content rather than performing simple pattern matching on labeled score-text pairs. We provided Expert-Tuned GPT-4.1-mini and Base GPT-5-mini models with the following prompt:

“You are an expert rater of essay creativity. Return a single creativity percentile from 0 to 1. You may see reference essays; use them only to calibrate your sense of creativity. For example, after reading the reference essays, consider how much the target essay demonstrates novel ideas, unique perspectives, and imaginative expression relative to those essays, and rate it accordingly. Answer with only a float between 0 and 1, nothing else.

Reference Essay: {a_set_of_reference_essays}

Target essay: {target_essay}

Return only a single float between 0 and 1.”

We fitted separate regression models for each of the eight conditions, defined by the number of reference essays provided (up to 128), to test if increased example exposure in joint evaluation significantly altered the predictive weight of surface-level semantic diversity. As shown in Extended Data Fig. 6, the inclusion of up to 128 reference essays did not change the predictive weight of specific semantic levels in systematic ways. This result suggests that joint evaluation had no systematic effects on how word-, sentence-, and document-level metrics relate to creativity ratings, indicating that lexical bias is not a byproduct of single evaluation and is a robust characteristic of the model’s evaluative framework.

<i>Predictors</i>	Elite Private (<i>N</i> = 71,268)		Regional Private (<i>N</i> = 66,354)		Flagship Public (<i>N</i> = 154,194)		Selective Liberal-Arts (<i>N</i> = 21,069)	
	β	95% CI	β	95% CI	β	95% CI	β	95% CI
Total Effect	0.20***	0.18, 0.22	0.24***	0.22, 0.26	0.19***	0.18, 0.20	0.30***	0.27, 0.32
Indirect Effect	0.12***	0.11, 0.13	0.06***	0.05, 0.07	0.06***	0.05, 0.07	0.20***	0.18, 0.22
Direct Effect	0.08***	0.07, 0.09	0.18***	0.17, 0.20	0.13***	0.13, 0.14	0.09***	0.07, 0.12
Indirect (Word)	0.13***	0.13, 0.14	0.08***	0.07, 0.09	0.08***	0.07, 0.09	0.21***	0.20, 0.23
Indirect (Sentence)	-0.01***	-0.01, -0.00	-0.02***	-0.02, -0.02	-0.01***	-0.01, -0.01	-0.01***	-0.01, -0.00
Indirect (Document)	-0.01***	-0.01, -0.01	-0.00***	-0.00, -0.00	-0.01***	-0.01, -0.00	-0.00***	-0.01, -0.00
Contrast (Word vs. Others)	0.12***	0.11, 0.13	0.06***	0.05, 0.07	0.06***	0.05, 0.07	0.20***	0.18, 0.22

Supplementary Table 14.

Natural Experiments: Parallel Mediation Analyses of LLM Release Effects on Creativity

Ratings. Mediation models used complete cases for the three semantic diversity measures and the expert-tuned GPT-4.1-mini creativity rating. A total of 12 essays (0.02%) from the elite private university, 26 (0.04%) from the regional private university, 456 (0.29%) from the flagship public university, and 9 (0.04%) from the selective liberal-arts college were excluded from the mediation analyses. These exclusions were due primarily to missing automated creativity ratings caused by invalid model outputs outside the valid 0–1 rating range (elite private: 12; regional private: 20; flagship public: 455; selective liberal-arts: 9). Seven additional cases lacked sentence-level diversity scores because misplaced or omitted punctuation prevented reliable sentence segmentation (regional private: 6; flagship public: 1; see Supplementary Tables 3b–c). Word-level diversity and document-level distinctness were available for these cases. All path coefficients (direct, indirect, and total effects) and predefined contrasts were tested using bootstrapping with 5,000 resamples to obtain confidence intervals. CI = confidence interval. *** $p < .001$.

Controlled Experiment 3: Controlled Within-Subjects Experiment

Data. We recruited 224 participants from the elite private university, all of whom had prior experience writing college admissions essays. We removed incomplete or invalid responses ($n = 5$) and further excluded data from participants who failed attention checks ($n = 16$), resulting in a final sample of 203 participants. The data were collected in the 2024-2025 academic year, meaning that 46 Seniors (22.66%) and 54 Juniors (26.60%) wrote their essays before the ChatGPT release, while 72 Sophomores (35.47%) applied during the transitional period, and 31 Freshmen (15.27%) applied after ChatGPT. The average age was 20.00 years ($SD = 1.38$), and 157 participants identified themselves as female, 41 as male, 4 as nonbinary, and 1 as self-described. Participants self-identified their race as follows: 63.55% White or Caucasian ($n = 129$), 15.76% Asian ($n = 32$), 3.45% Black or African American ($n = 7$). Additionally, 0.49% identified as American Indian or Alaska Native ($n = 1$), and 0.49% as Native Hawaiian or Other Pacific Islander ($n = 1$). A further 15.27% ($n = 31$) selected “Other” or identified as multiracial, and 0.99% ($n = 2$) chose not to report. Separately, 14.29% ($n = 29$) identified as Hispanic/Latine. LLM Use Frequency was assessed with a survey item asking how often participants used ChatGPT and/or other LLMs for any purpose; 87.68% ($n = 178$) reported at least some prior use.

Materials and Procedure. Data collection was conducted online. After providing informed consent, participants were asked to upload the college admissions essays they had previously submitted in their college applications. Because data collection was at the same elite private university where one of our natural experiments occurred, the uploaded essays were written in response to the same prompt used in the natural experiment. Participants were informed that they could redact any sensitive information before uploading. Next, participants were asked to write a detailed prompt (more than 150 words) that they would give to an AI to generate a new customized personal statement essay from scratch. Then, they were asked to

write another detailed prompt that they would use to instruct an AI to revise or improve their original essay, which they had originally submitted. Finally, participants completed questionnaires on demographics and creativity. All survey questionnaires are available on the OSF at <https://osf.io/yd94z>.

5 **Instructions for AI-Assisted Essay Generation Task.** Students received the following instructions:

10 *“Imagine you are applying to college. Your task is to craft a prompt to ChatGPT to write you a college admissions essay to OOOO University. Your goal is to prompt ChatGPT to create an essay that describes you, not a generic admissions essay. You must therefore provide personal details so that ChatGPT can create an essay specific to you. The essay must be around 650 words. Your prompt must contain at least 1000 characters (~150 words). You should write as much as you need to get the best results.*

15 *Now, please use the space below to draft a ChatGPT prompt to make the AI help you **create** a college admissions essay in response to the following essay question: ‘As OOOO University is a diverse community, the Admissions Committee would like to know more about you in your words. Please submit a brief essay, either personal or creative, which you feel best describes you.’ You may copy and paste this basic essay question, but you must also provide personal details to help ChatGPT create the best personalized admissions essay.”*

20 **Instructions for AI-Assisted Revision Task.** Students received the following instructions:

25 *“Imagine the situation again when you are applying to college. But, this time, you are using ChatGPT not to generate a whole new essay but to revise or improve the admissions essay you originally wrote. Your goal is to prompt ChatGPT to **revise** or improve your own essay you originally submitted to OOOO University. You must therefore provide personal details so that ChatGPT can edit your essay specific to you. The essay must be around 650 words. Your prompt must contain at least 1000 characters (~150 words). You should write as much as you need to get the best results.*

30 *Now, please use the space below to write a ChatGPT prompt to make ChatGPT help you revise or improve your college admissions essay. Please remember that you are responding to the following essay prompt: ‘As OOOO University is a diverse community, the Admissions Committee would like to know more about you in your words. Please submit a brief essay, either personal or creative, which you feel best describes you.’ You may copy and paste this basic essay question, but you must also provide personal details to help ChatGPT revise or improve your admissions essay for you.”*

Following data collection, we generated the two AI-generated essays for each participant. This step was crucial to ensure that all AI essays were generated using the same model and settings, controlling for technical variability. All AI content was generated using OpenAI's gpt-4.1-2025-04-14 model via its API, which was the most recent version available at the time of analysis. For the GPT-generated condition, each participant's individual generation prompt was fed into the model to produce a new customized essay. For the GPT-revised condition, each participant's original essay and their individual revision prompt were fed into the model to produce a revised version. This procedure resulted in a final dataset where each participant was associated with three distinct essays, allowing us to assess within-person effects of AI use.

To isolate the causal impact of AI use on semantic structure, we conducted a within-subjects experiment examining how semantic metrics at each level varied across authorship conditions: original essay, AI-revised, and AI-generated. We employed multilevel linear mixed-effects models for each semantic level, incorporating a random intercept for each participant to account for repeated measures across authorship conditions. We assessed differences between the three writing conditions using post-hoc pairwise comparisons, applying the Holm correction to control the family-wise error rate.

We further tested whether authorship condition effects were moderated by LLM Use Frequency. We entered LLM Use Frequency as a mean-centered continuous moderator and re-estimated the mixed-effects models with Condition $\times$ LLM Use Frequency interactions. These interactions were not significant at any semantic level, indicating that the effects of authorship condition did not depend detectably on participants' LLM use frequency (Supplementary Table 15b).

<i>Predictors</i>	Word-Level		Sentence-Level		Document-Level	
	β	95% CI	β	95% CI	β	95% CI
GPT-Revised Essay	0.79***	0.64, 0.94	-0.47***	-0.62, -0.31	-0.98***	-1.10, -0.85
GPT-Generated Essay	0.65***	0.50, 0.79	-0.50***	-0.66, -0.35	-1.52***	-1.64, -1.40
Random Effects						
σ^2	0.57		0.64		0.40	
τ_{00}	0.31		0.31		0.21	
<i>ICC</i>	0.35		0.32		0.34	
$N_{participant}$	203		203		203	
<i>Observations</i>	609		609		609	
Marginal R^2 / Conditional R^2	.118 / .428		.052 / .360		.396 / .602	

Supplementary Table 15a.

Controlled Experiment 3: Semantic Diversity by Authorship Condition Across Embedding Levels (Within-Subject). The reference group was the Human Essays. CI = confidence interval.

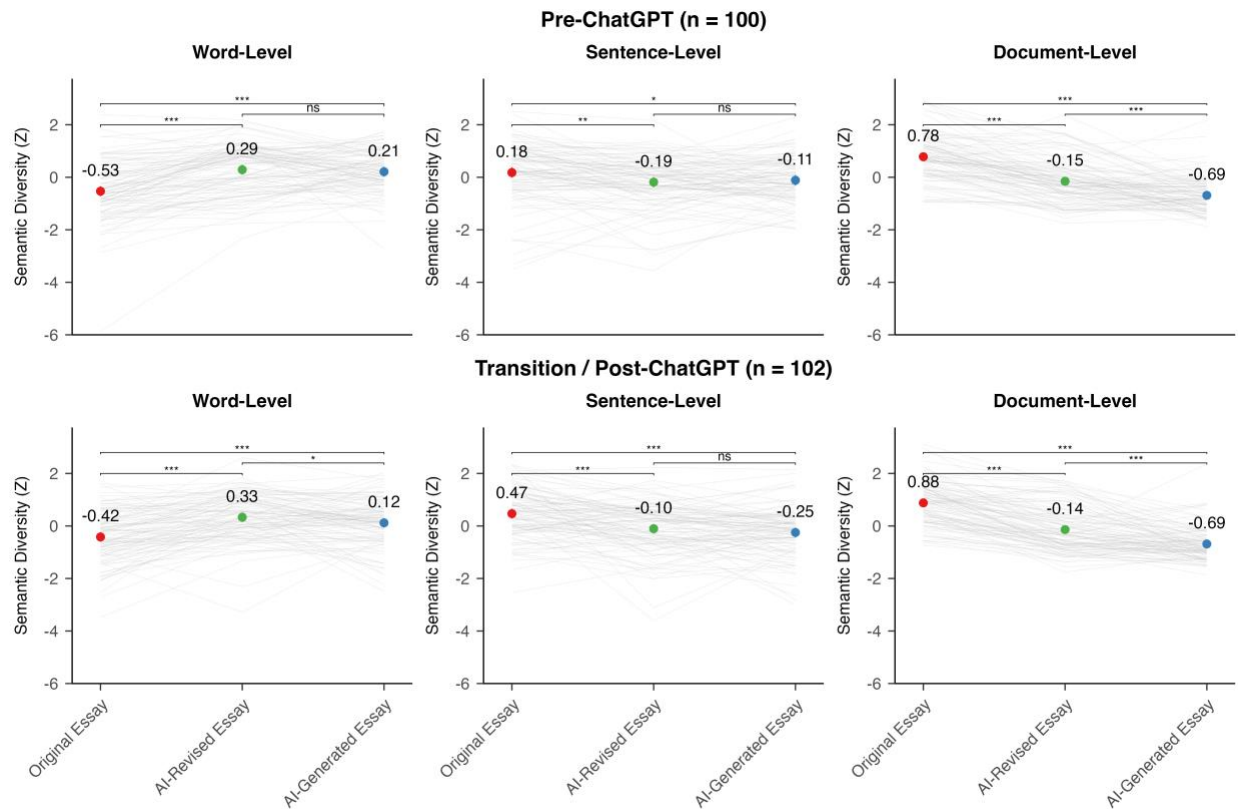
*** $p < .001$.

<i>Predictors</i>	Word-Level		Sentence-Level		Document-Level	
	β	95% CI	β	95% CI	β	95% CI
GPT-Revised Essay	0.79***	0.64 – 0.94	-0.47***	-0.62 – -0.31	-0.98***	-1.10 – -0.85
GPT-Generated Essay	0.65***	0.50 – 0.79	-0.50***	-0.66 – -0.35	-1.52***	-1.64 – -1.40
LLM Use Frequency	-0.05	-0.18 – 0.08	-0.02	-0.16 – 0.11	-0.07	-0.18 – 0.04
GPT-Revised Essay× LLM Use Frequency	0.06	-0.09 – 0.20	0.06	-0.10 – 0.22	0.07	-0.05 – 0.19
GPT-Generated Essay× LLM Use Frequency	-0.03	-0.18 – 0.12	-0.04	-0.19 – 0.12	0.07	-0.06 – 0.19
Random Effects						
σ^2		0.57		0.64		0.40
τ_{00}		0.31		0.31		0.21
<i>ICC</i>		0.35		0.33		0.34
$N_{participant}$		203		203		203
Observations		609		609		609
Marginal R^2 / Conditional R^2		.120 / .430		.054 / .362		.396 / .602

Supplementary Table 15b.

Controlled Experiment 3: Semantic Diversity by Authorship Condition and LLM Use Frequency Across Embedding Levels (Within-Subject). The reference group was the Human Essays. CI = confidence interval. *** $p < .001$.

To further isolate the impact of AI use, we repeated the same within-subjects models separately for participants whose original admissions essays were submitted before ChatGPT's public release, and for those whose essays were submitted either during the transition year (2023; some essays submitted this year were likely written before ChatGPT's release) or in 2024. For these analyses, we excluded one participant in the transition/post-ChatGPT group whose original essay was flagged by an AI detector (GPTZero) as likely AI-generated. The semantic disjunction pattern was replicated in both subsets, with AI-revised and AI-generated essays showing higher word-level diversity but lower sentence-level diversity and document-level distinctness than participants' original essays (Supplementary Fig. 1 and Supplementary Table 15c).



Supplementary Fig. 1.

Controlled Experiment 3: Within-Subject Effects of AI Use on Semantic Diversity Scores by Original Essay Submission Period. Participants were grouped according to whether their original admissions essays were submitted before ChatGPT’s public release or during/after the transition period. One participant in the transition/post-ChatGPT group, whose original essay was flagged as likely AI-generated, was excluded from the analyses. Dots and error bars show estimated marginal means and 95% confidence intervals from mixed-effects models for each authorship condition (Supplementary Table 15c). Post-hoc pairwise comparisons were used to compare estimated marginal means with Holm correction for multiple comparisons, conducted separately for each semantic level and original essay submission period. $*p < .05$, $**p < .01$, $***p < .001$.

<i>Predictors</i>	Pre-ChatGPT			Transitional / post-ChatGPT		
	Word-Level	Sentence-Level	Document-Level	Word-Level	Sentence-Level	Document-Level
	β	β	β	β	β	β
GPT-Revised Essay	0.80***	-0.36**	-0.94***	0.77***	-0.57***	-1.01***
GPT-Generated Essay	0.72***	-0.29**	-1.48***	0.55***	-0.72***	-1.55***
Random Effects						
σ^2	0.57	0.63	0.42	0.57	0.63	0.39
τ_{00}	0.35	0.35	0.21	0.28	0.27	0.21
<i>ICC</i>	0.38	0.36	0.33	0.33	0.30	0.35
$N_{participant}$	100	100	100	102	102	102
<i>Observations</i>	300	300	300	306	306	306
Marginal R^2 / Conditional R^2	.129 / .459	.025 / .373	.373 / .581	.105 / .400	.096 / .370	.414 / .620

Supplementary Table 15c.

Controlled Experiment 3: Semantic Diversity by Authorship Condition Across Embedding Levels and Original Essay Submission Period (Within-Subject). Models were estimated separately for participants whose original admissions essays were submitted before ChatGPT's public release and for those whose essays were submitted during/after the transition period. Models were estimated after excluding one participant in the transition/post-ChatGPT group whose original essay was flagged as likely AI-generated. The reference group was the Human Essays. * $p < .05$, ** $p < .01$, *** $p < .001$.